

Targeted Distraction: Notifications, Productivity, and Platform Regulation*

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Abstract

Mobile platforms use push notifications to generate timely engagement, but these prompts can arrive when attention has a high opportunity cost. I study app-mediated food delivery, where smartphones are both production tools and sources of distracting prompts. One additional interruption raises average trip duration by 4.8 percent. Eliminating all interruptions reduces trip duration by 12.0 percent and raises output by \$1.43 per hour. The burden is heterogeneous: young workers receive more social-media prompts, while men face larger production losses. Counterfactuals show that muting, batching, and depersonalizing notifications recover meaningful productivity gains by changing when and to whom platforms prompt.

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1 Introduction

Mobile platforms increasingly decide not only what users see, but also when to demand their attention. Push notifications are a central instrument in this design problem. They can be personalized, timed, and delivered to a device that is nearly always within reach. A large literature shows that targeting, personalization, and mobile context can make digital messages more effective by reaching users at moments when they are likely to respond (Goldfarb and Tucker, 2011; Lambrecht and Tucker, 2013; Luo, Andrews, Fang, and Phang, 2014; Tucker, 2014; Bleier and Eisenbeiss, 2015; Andrews, Luo, Fang, and Ghose, 2016; Fong, Fang, and Luo, 2015). The same feature that makes a prompt effective, however, can make it costly in other contexts. When a notification arrives during work, the response it induces is also an interruption: attention shifts away from production, output may be delayed, and the value of the message depends on more than the click it generates.

This paper studies the productivity cost of notification design. The policy relevance is immediate. Recent legal and regulatory debates over platform design increasingly focus not only on user choices, but also on how digital products shape the opportunity set users face. Notification policy is one concrete margin in this debate. Demand-side tools, such as do-not-disturb rules or app-specific muting, let users block prompts during high-opportunity-cost periods. Supply-side rules, such as batching, caps, or limits on personalization, change the stream of prompts before users decide whether to respond. Evaluating these policies requires three objects that are rarely observed together: the supply of notifications, users’ interruption choices, and the productivity consequences of those interruptions.

I study these objects in the setting of app-mediated food delivery. The setting is useful because the smartphone is both a production tool and a source of non-work prompts. Couriers need the phone for delivery and navigation, but the same device delivers social-media and communication notifications throughout the work spell. Output is also measured at high frequency: delivery work consists of repeated trips with observable timestamps, distances, and durations. This makes it possible to connect notification exposure, interruption choices, and realized productivity within the same work environment.

The empirical analysis combines high-frequency Android panel data from Global Wireless Solutions (GWS) and an auxiliary field experiment from Bao and Imas (2026) that records delivered notifications. The GWS panel records app-level foreground episodes, timestamps, locations, and demographics for U.S. Android users from January 2019 through June 2024. I identify food-delivery couriers through driver-facing delivery apps, construct work spells and delivery trips from app activity and GPS movement, and measure non-work interruptions during both idle intervals and delivery trips. The field experiment provides a direct mapping between delivered notifications and subsequent app openings, which I use to calibrate notification click-through rates.

The first part of the paper estimates the supply of push notifications. Because the GWS panel does not directly record delivered notifications, I infer notification-induced openings from screen-session initiators. A direct transition into app a after an off-screen idle gap, when a was not the previously active app, is the transition most consistent with an actionable prompt from app a . I com-

bine these imputed openings with exposure time, estimate app-specific Poisson arrival processes, and use the field-experiment calibration to recover latent push-arrival rates for social-media and communication apps.

The estimated notification environment is large and highly personalized. In the post-2021 estimation period, social-media apps generate an average of 46.0 push notifications per user-day, and communication apps generate 45.0. Within each app, the personalized component is strongly right-skewed: most users receive below-average notification intensity, while a smaller group receives substantially higher intensity. This personalization is systematically related to demographics. Averaging across major social-media apps, users aged 18–29 have personalized push multipliers of 1.42, compared with 0.55 for users aged 40 and above. Communication apps display the opposite age profile: older users receive relatively more communication notifications. These patterns imply that personalization changes not only the level of notification exposure, but also its incidence across users.

The second part of the paper estimates the production cost of interruptions. I use a trip-level production function in which delivery duration per mile depends on interruptions per mile, trip distance, cumulative spell time, weather, pay, and fixed effects. The main identification concern is that slow trips may themselves create opportunities for phone use. I therefore instrument interruptions using app-side shift-share shocks and the worker’s lagged interruption rate. The shift-share instrument combines leave-one-out app-level interruption shocks with predetermined worker exposure shares, isolating variation in the salience of apps a worker was likely to use.

The estimates imply that interruptions are costly. One additional interruption on the average delivery trip raises trip duration by 4.8 percent, or about 1.2 minutes. The effect is tied to discrete attention shifts rather than total screen exposure: controlling for non-work screen time does not remove the interruption effect. The mechanism is also informative. Interruptions increase stop time, and couriers partly compensate by moving faster during the travel portion of the trip. The delay channel dominates, so total trip duration rises. The burden is uneven: one additional interruption raises duration by 5.3 percent for couriers aged 18–29, compared with 4.6 percent for couriers aged 40 or older, and by 5.3 percent for men, compared with 4.3 percent for women.

The third part of the paper estimates a dynamic model of interruption choices. The model distinguishes externally triggered interruptions, which are constrained by realized notification supply, from internally initiated checks. Workers receive utility from phone activity, but interruptions during delivery trips lengthen the current trip, raise cumulative spell time, reduce continuation probabilities, and lower expected output. Identification comes from comparing phone-use choices in idle periods and delivery trips. Idle periods reveal the static utility of phone activity because output is zero and idle duration is exogenous. Delivery trips reveal the worker’s revealed-preference value of output because the same phone activity now carries dynamic production costs.

The model delivers three behavioral primitives that matter for policy. First, external prompts carry positive premia relative to otherwise similar self-initiated checks, so notifications do more than reveal latent demand for an app; they raise the utility of immediate engagement. Second, interruptions cluster within category: one social-media or communication prompt can initiate

additional checking rather than a single isolated response. Third, app categories substitute for one another within a period, implying that restricting one category of notifications need not reduce total phone activity one-for-one. These margins determine how workers respond to muting, batching, and depersonalizing notifications.

I then use the estimated model to evaluate notification policies. The counterfactuals separate an upper-bound benchmark that removes all non-work interruptions from feasible policies that operate on external notifications. Eliminating all interruptions reduces delivery-trip duration by 12.0 percent on average, or about 3.8 minutes per trip, and raises output by \$1.43 per hour. External notifications explain an important, but incomplete, share of this cost. Muting both social-media and communication notifications reduces trip duration by 2.3 percent and raises income by \$0.27 per hour. Muting either category alone has about half that effect. Batching both categories at a 30-minute frequency reduces trip duration by 1.5 percent and raises income by \$0.16 per hour, preserving some notification flow while limiting bursts. Removing social-media personalization has a smaller average effect, but its gains are concentrated among highly targeted workers.

The distributional effects sharpen the policy lesson. Social-media restrictions disproportionately benefit young workers, who receive more social-media notifications. Communication restrictions are more balanced across age groups and are especially relevant for older men. Broad muting or batching policies combine these margins and generate the largest feasible gains, while depersonalization is narrower and reallocates notification exposure away from highly targeted workers. These results do not imply that all notifications should be eliminated. Communication prompts may have coordination value, and workers may value internally initiated phone activity. The estimates instead quantify the productivity side of a design trade-off: mobile prompts create engagement value, but they impose opportunity costs when they reach users in high-value production contexts.

Related literature. The paper contributes to three literatures. First, it contributes to research on digital advertising, targeting, personalization, and mobile marketing. One strand studies online advertising markets and measurement, including sponsored-search auctions, banner advertising, paid-search spillovers, and advertising spillovers ([Manchanda, Dubé, Goh, and Chintagunta, 2006](#); [Edelman, Ostrovsky, and Schwarz, 2007](#); [Rutz and Bucklin, 2011](#); [Sahni, 2016](#)). A second strand studies how targeting and personalization affect advertising and recommendation effectiveness, including display-ad targeting, retargeting, personalized social advertising, and personalized online banners ([Goldfarb and Tucker, 2011](#); [Lambrecht and Tucker, 2013](#); [Tucker, 2014](#); [Bleier and Eisenbeiss, 2015](#)). A third strand studies mobile products and mobile promotion design, showing how app design, product type, time, location, competitive proximity, and physical context shape response to mobile messages ([Ghose and Han, 2014](#); [Bart, Stephen, and Sarvary, 2014](#); [Luo, Andrews, Fang, and Phang, 2014](#); [Fong, Fang, and Luo, 2015](#); [Andrews, Luo, Fang, and Ghose, 2016](#)). These papers establish that data, timing, and context can increase response. My contribution is to measure a different object: the downstream cost of effective prompting when the response occurs in a high-opportunity-cost state. The paper therefore moves the marketing-

design margin from response and conversion to the joint problem of engagement, interruption, and output. It also provides a measurement framework for a key object in mobile marketing—latent push-notification supply—and estimates how personalization changes both the level and incidence of exposure across users.

Second, it contributes to work on digital addiction, attention, and self-control. The theoretical foundation comes from models in which the decision maker may value an action differently at the moment of choice than from an *ex ante* or long-run perspective (Strotz, 1955; Loewenstein and Thaler, 1989; Thaler and Shefrin, 1981; Laibson, 1997; O’Donoghue and Rabin, 1999; Gul and Pesendorfer, 2001; Fudenberg and Levine, 2006). A related economics literature models addictive consumption and commitment problems (Becker and Murphy, 1988; Gruber and Köszegi, 2001). Empirical evidence in marketing and economics studies self-control in consumption, commitment, tempting goods, and work settings (Hoch and Loewenstein, 1991; Wertenbroch, 1998; DellaVigna and Malmendier, 2006; Kaur, Kremer, and Mullainathan, 2015; Schilbach, 2019). Recent work applies these ideas to digital products and social media, estimating welfare effects, addiction-like use, and demand for commitment devices (Allcott, Braghieri, Eichmeyer, and Gentzkow, 2020; Allcott, Gentzkow, and Song, 2022; Bao, 2025; Bao and Imas, 2026). My contribution is to shift the unit of analysis from platform time or platform valuation to the notification-to-interruption channel inside production. I distinguish externally triggered interruptions from internally initiated checks, estimate the prompt premium attached to notifications, and use observed choices across idle and delivery periods to recover the revealed-preference value of output. This lets the paper quantify how a behavioral or social-cost wedge would translate into welfare through a sufficient statistic: lost productivity.

Third, it contributes to research on platform work, alternative work arrangements, and productivity measurement. A labor literature studies the rise and value of flexible or alternative work arrangements (Mas and Pallais, 2017; Katz and Krueger, 2019); a related platform-work literature studies worker participation, flexibility, matching, platform efficiency, pay, and inequality on ride-hailing and peer-to-peer platforms (Cramer and Krueger, 2016; Chen, Chevalier, Rossi, and Oehlsen, 2019; Hall and Krueger, 2018; Cullen and Farronato, 2021; Cook, Diamond, Hall, List, and Oyer, 2021). My contribution is to bring the platform-design margin into this labor setting. Rather than treating the smartphone only as a matching or monitoring technology, I study it as both the production tool and the source of non-work prompts. I measure discrete interruptions during delivery trips, use app-side shocks to identify their causal production cost, and embed that cost in a dynamic model of work spells. The resulting counterfactuals show that notification supply is not merely a product-design choice for consumer platforms; it is also a labor-market margin that changes output, earnings, and the incidence of productivity losses across workers.

The remainder of the paper is organized as follows. Section 2 describes the data and empirical setting. Section 3 develops the structural framework and the welfare interpretation. Section 4, 5, and 6 estimate notification supply, the production cost of interruptions, and interruption preferences, respectively. Section 7 evaluates notification policies, and Section 8 concludes.

2 Data and Empirical Setting

This section introduces the data sources and measurement objects used throughout the paper. The empirical analysis has three layers. First, I use a broad phone-use panel to measure app activity and estimate notification supply. Second, I restrict attention to food-delivery couriers to measure delivery output and the production cost of interruptions. Third, I combine these objects in a matched structural panel that contains notification-arrival inputs, phone-use choices, delivery productivity, payoff incentives, weather controls, and demographics. I describe the data sources first, then define the common objects that link the notification, production, and structural analyses.

2.1 Data sources

The primary data source is Global Wireless Solutions (GWS) Magnify, accessed through the Dewey Data platform. The data consist of an anonymized, opt-in panel of Android users in the United States and record high-frequency app-level screen activity. For each foreground-app episode, I observe the app name, start and end timestamps, GPS coordinates, device model, and user demographics, including age, gender, and ethnicity. The sample spans January 2019 through June 2024 and includes 607,591 users, who are observed for 118 days on average. These records are the basis for the paper’s three core measurements: app openings used to infer notification supply, delivery trips and idle periods for food-delivery couriers, and non-work phone interruptions during working spells. [Appendix A.1](#) provides additional details

I complement the GWS panel with data from a field experiment conducted by [Bao and Imas \(2026\)](#). Participants installed an Android logging app that records screen activity, foreground app use, and the timestamp and source app of each posted push notification. Unlike the GWS panel, the experimental data directly observe delivered notifications. I use these records to validate the notification-imputation procedure and to calibrate the probability that a delivered push notification generates an app opening that is observed by the GWS-based rule.

I also merge external data to construct payoffs and local controls. For courier earnings, I anchor per-mile delivery compensation using New York City Department of Consumer and Worker Protection data on restaurant-delivery workers, focusing on car couriers, and extrapolate this benchmark across states and quarters using the Bureau of Labor Statistics Quarterly Census of Employment and Wages. For operating costs, I combine Energy Information Administration gasoline prices with Federal Highway Administration fuel-economy information to construct gasoline cost per mile. The payoff measure used in the production and structural analyses is net payoff per mile: gross delivery compensation per mile net of gasoline cost. Finally, I merge state-day weather controls from NOAA nClimGrid-Daily, specifically average temperature and precipitation, because weather affects travel conditions and the opportunity cost of continuing a delivery spell.

Throughout the paper, I partition users into six demographic groups,

$$g \in \mathcal{G} = \{\text{female, male}\} \times \{18\text{--}29, 30\text{--}39, 40+\}.$$

This grouping balances heterogeneity and precision. Age and gender are observed in the GWS panel and are central to the paper’s empirical margins: notification supply, interruption costs, phone-use preferences, and counterfactual gains all vary along these dimensions. At the same time, the group-specific production and structural estimates require a food-delivery courier sample with matched notification inputs, production inputs, payoff controls, weather controls, and observed interruption choices. The final structural sample contains all six cells, with at least 620 workers and 17,543 working spells in each cell. Finer age bins, or cross-classification by ethnicity, would leave some cells too small for the group-specific production functions and dynamic preference estimates.

2.2 Empirical architecture

The paper uses different restrictions of the same data environment. Section 4 uses the broad phone-use panel to estimate latent push-notification arrival rates for social-media and communication apps. Section 5 restricts attention to food-delivery couriers and estimates how interruptions affect delivery-trip duration. Section 6 uses the matched structural panel, in which phone-use choices, notification-arrival inputs, production inputs, payoff controls, weather controls, and demographic groups are observed jointly. I therefore report sample-specific summary statistics in the empirical sections where each sample is constructed. This subsection defines the common objects that link the exercises.

The central phone-use measures are notifications, interruptions, and screen duration. First, the GWS panel does not directly record notifications; however, Section 4 shows that app-use patterns can be used to estimate notification arrival rates at the user–app–day level and then map those rates into worker–spell arrival rates. I estimate this supply for two policy-relevant app families: social media (SM) and communication (CM). Second, I define a phone interruption as either (i) a transition from off-screen to on-screen activity that opens an app, or (ii) for workers, a switch from a work-related app to a non-work app. Interruptions may be externally triggered by notifications or internally initiated by the user; the primary GWS data records only the total number of interruptions, which combines both sources. Third, screen duration is the total time spent on apps within a given time interval. I construct interruption and duration measures at the app level and aggregate non-work apps into three categories used throughout the model: social media (SM), communication (CM), and other apps (OT). Only SM and CM have modeled external notification arrivals; OT interruptions remain in the production and preference analyses as internally initiated non-work phone activity.

A second key object is productivity. Unlike phone-use measures, which are defined uniformly across the panel, productivity requires occupation-specific output measures and is therefore available only for selected workers in the GWS sample. I focus on *food-delivery couriers*, whose work activity can be inferred from dedicated courier apps and GPS traces. I identify couriers through usage of “Dasher” and “Grubhub for Drivers” and construct *work spells* as contiguous intervals

of delivery-app activity, during which the courier is logged on to receive delivery offers.¹ Within each spell, GPS traces partition time into alternating *delivery trips* and *idle periods*, matching the timing structure of the framework in Section 3. For each trip, I observe distance and duration, which provide high-frequency measures of delivery output and productivity under distance-based compensation. Work-related apps include delivery, navigation, and task-completion apps: “Dasher,” “Grubhub for Drivers,” “Google Maps,” “Waze,” and “Camera.” All other foreground episodes are classified into the non-work categories above. Section 5 describes the spell and trip construction, sample restrictions, and summary statistics in detail.

3 Structural Framework

Building on the measurement objects introduced above, I develop a dynamic model of a delivery worker who allocates attention between production and phone activity over the course of a working spell. The model formalizes three links that are central for policy: notification supply shifts the worker’s feasible set of interruptions; interruptions affect production by changing trip duration and future fatigue; and the resulting welfare shadow prices can be aggregated to evaluate supply-side notification regulation. I keep the framework general and estimate its key components in the empirical sections that follow.

3.1 Setup

Environment. I focus on a single worker i during a working spell s and omit worker–spell subscripts when they are unambiguous. Periods $t = 1, 2, \dots$ alternate between *delivery trips* in odd periods and *idle intervals* in even periods. Each period is summarized by its duration ℓ_t , distance m_t , and cumulative spell time $L_t = \sum_{\tau < t} \ell_\tau$. For delivery trips, distance m_t is assigned by the platform, and duration $\ell_t = m_t \Delta(X_t, m_t, L_t)$ depends on the number of interruptions X_t , distance m_t , fatigue measured by L_t , and richer spell-level controls which I introduce in estimation. For idle intervals, $m_t = 0$ and ℓ_t is drawn exogenously.

Let app categories be indexed by $c \in \mathcal{M} = \{SM, CM, OT\}$, corresponding to social media, communication, and other apps. Let $\mathcal{M}^* = \{SM, CM\}$ denote the subset of app categories that generate most notifications.² The worker is exposed to two sources of phone activity. *External* notifications arrive from social-media and communication apps at worker–spell-specific rates $\{\lambda_c\}_{c \in \mathcal{M}^*}$; the realized notification vector in period t is $\mathbf{n}_t = (n_{ct})_{c \in \mathcal{M}^*}$. *Internal* interruptions are self-initiated checks of apps in each category.

The worker’s action is $\mathbf{x}_t = (\mathbf{x}_t^{ex}, \mathbf{x}_t^{in})$. The external component, $\mathbf{x}_t^{ex} = (x_{ct}^{ex})_{c \in \mathcal{M}^*}$, specifies how many incoming notifications the worker acts on in each notification-generating category, subject to $x_{ct}^{ex} \leq n_{ct}$. The internal component, $\mathbf{x}_t^{in} = (x_{ct}^{in})_{c \in \mathcal{M}}$, specifies how many self-initiated checks the

¹I exclude UberEats couriers because the Uber Driver app combines ride-hailing and food-delivery work.

²I focus on notifications generated by social-media and communication apps, which are central to current policy debates over attention, addiction, and platform design.

worker makes in each app category. Total interruptions are

$$X(\mathbf{x}_t) = \sum_{c \in \mathcal{M}^*} x_{ct}^{ex} + \sum_{c \in \mathcal{M}} x_{ct}^{in}. \quad (1)$$

Let $\mathcal{X}(\mathbf{n}_t)$ denote the corresponding feasible action set. The constraints $x_{ct}^{ex} \leq n_{ct}$ are the channel through which platform notification policy enters the worker's problem, and their shadow prices are the central objects for policy evaluation.

Each period generates two payoff components. First, the worker receives utility $u(\mathbf{x}_t, \ell_t)$ from phone activity. I assume that this utility is homogeneous of degree one in period duration, so phone utility scales linearly with the length of the period.³ The specification will be flexible to capture substitution between external and internal interruptions, as well as substitution across app categories. Second, the worker receives a production payoff $\alpha\omega m_t$, where ω is the per-mile wage that could vary across spells and α converts dollars into utils.

After completing a delivery trip t , the worker continues the spell with probability $\rho(L_t + \ell_t)$, where $L_t + \ell_t$ is cumulative spell duration at the end of the trip. I assume that $\rho(\cdot)$ is decreasing in spell duration, so longer spells are more likely to terminate. This monotonicity provides a reduced-form way to capture fatigue while leaving the stopping process otherwise exogenous. In estimation, I allow this probability to also depend on the richer observed state.

Dynamic problem. Let $EW(L)$ and $EV(L)$ denote the ex ante value of entering an idle interval and a delivery trip, respectively, with cumulative spell duration L as the only state variable. The worker faces two decisions along each idle-delivery cycle.

First, upon entering an idle period and observing $(\ell, \mathbf{n})$, the worker chooses an action $\mathbf{x} \in \mathcal{X}(\mathbf{n})$. The realized value of the idle period is

$$W(L; \ell, \mathbf{n}, \epsilon) = \max_{\mathbf{x} \in \mathcal{X}(\mathbf{n})} \left\{ u(\mathbf{x}, \ell) + EV(L + \ell) + \epsilon_{\mathbf{x}} \right\}, \quad (2)$$

where ϵ are action-specific shocks.⁴ The ex ante value of entering an idle period with cumulative spell time L is therefore

$$EW(L) = \mathbb{E}_{\ell, \mathbf{n}, \epsilon} [W(L; \ell, \mathbf{n}, \epsilon)]. \quad (3)$$

Because the worker is not productive during idle periods and idle duration ℓ is exogenous, $EV(L + \ell)$ is fixed at the time of the phone-use decision. The idle-period choice therefore maximizes the static payoff, which provides a tractable source of identification for the preference function $u(\cdot)$.

After the idle period, the worker starts the next delivery trip with cumulative spell time L' and observes $(m', \mathbf{n}')$. The worker chooses an action $\mathbf{x} \in \mathcal{X}(\mathbf{n}')$ according to

$$V(L'; m', \mathbf{n}', \epsilon') = \max_{\mathbf{x} \in \mathcal{X}(\mathbf{n}')} \left\{ u(\mathbf{x}, \ell'_{\mathbf{x}}) + \alpha\omega m' + \rho'_{\mathbf{x}} EW(L' + \ell'_{\mathbf{x}}) + \epsilon'_{\mathbf{x}} \right\}, \quad (4)$$

³This homogeneity assumption allows utility to be aggregated across periods of different lengths.

⁴I use semicolons to separate state variables from static latent shocks.

where

$$\ell'_x \equiv m' \Delta(X(\mathbf{x}), m', L') \quad \text{and} \quad \rho'_x \equiv \rho(L' + \ell'_x)$$

are the endogenous delivery duration and continuation probability implied by action $\mathbf{x}$, respectively. The continuation probability ρ'_x is decreasing in completed spell duration and serves as the effective discount factor. It therefore captures the fact that actions that lengthen the current trip raise cumulative spell time and reduce the probability that the worker remains active. The resulting ex ante value of starting a delivery trip is

$$EV(L') = \mathbb{E}_{m', \mathbf{n}', \epsilon'} [V(L'; m', \mathbf{n}', \epsilon')]. \quad (5)$$

The Bellman recursions in equations (2)–(5) characterize the value of entering an idle period, $EW(L)$, and the value of starting a delivery trip, $EV(L)$. This recursive structure makes interruptions *dynamically* costly: an additional check lengthens the current delivery trip, raises cumulative spell time, lowers the probability that the worker continues to work, and reduces the expected number of trips completed during the spell. This is the channel through which a single notification can generate first-order production losses.

Empirical Roadmap. The framework decomposes into four pieces that the rest of the paper estimates in turn. Section 4 estimates the personalized notification arrival process, recovering $\hat{\lambda}_{is}$ and documenting how it varies systematically across individuals and over time. Section 5 estimates the production-duration function $\hat{\Delta}_{is}(X, m, L)$ from a shift-share IV strategy that delivers the causal effect of interruptions on trip duration. Section 6 estimates the remaining preference parameters from the worker’s interruption choices. Section 7 feeds these objects into a set of demand- and supply-side counterfactuals, including do-not-disturb rules, notification caps, and bans on personalized notification, and reports their welfare consequences.

3.2 Welfare interpretation

The structural model is a revealed-preference benchmark. The worker maximizes expected utility taking the notification supply as given, and the coefficient identified from observational choices is the rate at which the worker trades off phone-use utility against monetary output while on the job. I denote this revealed-preference valuation of output by α^{doer} . Under this benchmark, no behavioral wedge is built into the choice problem: conditional on the observed notification environment, the chosen allocation is optimal for the worker’s revealed preferences.

Inefficiency. A long literature on self-control problems nevertheless suggests that this revealed-preference valuation may understate how much the same worker would value output under ex ante commitment. Hyperbolic-discounting and dual-self models of intertemporal choice predict precisely such a gap (e.g., [Strotz, 1955](#); [Thaler and Shefrin, 1981](#); [Laibson, 1997](#); [O’Donoghue and Rabin, 1999](#); [Gul and Pesendorfer, 2001](#); [Fudenberg and Levine, 2006](#)). Recent empirical work on

smartphone, social-media, and short-form-content use provides evidence consistent with this interpretation: users exhibit demand for commitment or self-control devices, and experimentally reducing exposure can improve subjective well-being or reduce subsequent valuations of the platform (e.g., Allcott, Braghieri, Eichmeyer, and Gentzkow, 2020; Allcott, Gentzkow, and Song, 2022; Bao, 2025; Bao and Imas, 2026). Mapping this evidence into the present model, the worker's *doer* self that chooses interruptions during the spell carries coefficient α^{doer} , while the *planner* self that would choose their own workplace policy ex ante carries coefficient $\alpha^{\text{planner}} \geq \alpha^{\text{doer}}$. The resulting wedge,

$$\tau^{\text{planner}} \equiv \alpha^{\text{planner}} - \alpha^{\text{doer}} \geq 0, \quad (6)$$

makes interruptions excessive relative to the planner's criterion and creates a welfare-relevant distortion in the planner's metric.

A regulator faces an additional consideration that neither self internalizes. To the extent that delays on a delivery trip are borne partly by consumers waiting for orders, or by other couriers through congestion or backlog effects, the social value of avoiding lost productivity exceeds the worker's private ex ante value. Internalizing these externalities raises the regulator's output coefficient above the planner's, so that $\alpha^{\text{regulator}} \geq \alpha^{\text{planner}} \geq \alpha^{\text{doer}}$. Equivalently, defining the regulator's wedge relative to the doer's revealed-preference benchmark as

$$\tau^{\text{regulator}} \equiv \alpha^{\text{regulator}} - \alpha^{\text{doer}} \geq \tau^{\text{planner}} \geq 0. \quad (7)$$

Each successive inequality strengthens the case for supply-side notification regulation relative to the rational revealed-preference benchmark.

Welfare metrics. The value function identified from observed choices is the doer's revealed-preference welfare. Policy evaluation instead requires evaluating the allocation induced by the doer's behavior under alternative welfare criteria. I index these criteria by a wedge τ , where $\tau = 0$ corresponds to the revealed-preference benchmark and $\tau > 0$ assigns additional value to output. Let $\alpha(\tau) \equiv \alpha^{\text{doer}} + \tau$, and let $\mathbf{x}^*(L, m, \mathbf{n}, \epsilon)$ denote the doer's policy rule from the delivery-trip problem in equation (4). For any wedge τ , define the shock-integrated welfare value of starting a delivery trip with cumulative spell time L , while holding fixed the doer's behavioral policy, as

$$\widetilde{EV}^\tau(L) = \mathbb{E}_{m, \mathbf{n}, \epsilon} \left[u(\mathbf{x}^*, \ell_{\mathbf{x}^*}) + (\alpha + \tau)\omega m + \rho_{\mathbf{x}^*} \widetilde{EW}^\tau(L + \ell_{\mathbf{x}^*}) + \epsilon_{\mathbf{x}^*} \right], \quad (8)$$

where $\rho_{\mathbf{x}^*} \equiv \rho(L + \ell_{\mathbf{x}^*})$ and the continuation value $\widetilde{EW}^\tau$ is evaluated using output weight $\alpha + \tau$, holding fixed the doer's subsequent phone-use and continuation policies.

This representation makes clear why productivity is the central welfare object. Let $M^*(L)$ denote the expected continuation-weighted miles completed from the beginning of a delivery trip with cumulative spell time L , including the current trip and all future trips, under the doer's behavioral policy. Because the only difference between welfare criterion τ and the revealed-

preference benchmark is the weight placed on output, the welfare value admits the decomposition

$$\widetilde{EV}^\tau(L) = \widetilde{EV}^0(L) + \tau\omega M^*(L). \quad (9)$$

Thus, expected output is a sufficient statistic for the welfare correction associated with any wedge τ . The empirical production function is therefore central to the welfare analysis: by mapping interruptions into trip duration and hence into expected miles completed per unit of working time, it determines the productivity component of the welfare cost of notifications.

4 Push Notification Supply

Notifications enter the worker’s attention problem as a supply-side constraint: they expand the set of externally triggered interruptions available to the worker. Measuring this supply is difficult because the GWS panel observes app openings, not delivered push notifications. This section constructs a measure of latent push-notification supply for social-media and communication apps. I first infer notification-induced openings from screen-session initiators, then estimate app-specific Poisson arrival models, and finally use external experimental data to map observed openings into latent push arrivals. The resulting estimates provide the app-specific notification arrival rates used in the structural model and quantify the personalized notification margins manipulated in the counterfactual policies.

4.1 Estimation strategy

I estimate notification-arrival processes for two policy-relevant app families: social media and communication. The social-media family includes social-network, social-messaging, and social-video apps such as Facebook Messenger, Facebook, YouTube, Instagram, TikTok, Snapchat, WhatsApp, Discord, Twitter (X), Telegram, and Reddit. The communication family includes phone, calling, messaging, calendar, and email utilities such as Phone, Call, Gmail, Messaging, Messages (Google), Google Duo, and Outlook.⁵ The analysis proceeds in three steps. First, I infer notification-induced openings from screen-session initiators in the GWS panel and combine them with app-specific exposure to construct a user-app-day arrival panel. Second, I estimate a Poisson model that decomposes opening rates into persistent individual-app heterogeneity, individual-app time trends, and app-day intensity. Third, I use external experimental evidence from [Bao and Imas \(2026\)](#) to calibrate effective click-through rates, mapping imputed notification-induced openings into latent push-notification supply.

Inferring notifications. The GWS panel records one row for each foreground-app episode, including home screens, app launchers, app switchers, and system UI, but does not directly observe

⁵The taxonomy follows the policy categories used in the counterfactuals rather than app-store labels. Some messaging apps are grouped with social media because they are embedded in social-network or social-platform environments.

push notifications. I therefore infer *notification-induced openings* from session initiators, defined as the first foreground app opened in a contiguous screen-on session. Consider the idle gap between two consecutive screen-on sessions. For a target app a , I treat the gap as exposure whenever the terminal app in the previous session is not a . If the next session begins with app a , I classify the transition as a notification-induced opening for a ; if the next session begins with another app, the gap contributes exposure but no opening for a .⁶

The rationale is that, absent an app-specific prompt, a new screen-on session should leave an observable trace of manual navigation. If the user simply resumes phone use, the first foreground app in the next session is typically the app that was active when the previous session ended. If instead the user manually switches to app a , the path usually passes through an observed foreground surface, such as the home screen, launcher, app switcher, or recents menu. In either case, app a would not appear as the first foreground episode immediately after the off-screen gap. A direct transition from a previous terminal app different from a to app a as the next session initiator is therefore the transition most consistent with an actionable prompt from app a , such as a lock-screen notification or a notification-shade tap.

This classification should still be interpreted as a proxy for notification-induced openings rather than as direct observation of push arrivals. It may still capture some manual opens through recents, widgets, shortcuts, or other entry points; these cases generate measurement error in the imputed opening count. To limit contamination from sleep and long off-phone spells, I drop idle gaps longer than one hour. [Appendix B.1](#) details the construction and reports validation exercises that benchmark the imputation. As a sanity check, the cross-app ranking of imputed notification-induced opening rates is intuitive: messaging and email apps rank highest, social-media apps fall in the middle, content and search apps such as YouTube, Spotify, and Chrome rank lower, and negative controls such as offline games are near zero.

The resulting estimation panel records the imputed notification-induced opening count, exposure time, and the user’s demographics for each user–app–day. Exposure time is the total idle time for which the previous foreground session did not end with the focal app. The main estimation sample covers January 1, 2021 through June 27, 2024.⁷ Table 1 summarizes the sample for the largest apps in each family and pool the remaining retained apps into residual “Other” rows. The table also provides a useful check on sample composition. Facebook and Messenger users are older and more female than users of TikTok, Snapchat, and Instagram; YouTube users are comparatively more male-skewed; and the large communication-app samples are concentrated around ages 39–40, with female shares near 57 percent. These patterns are consistent with broad cross-platform audience differences while preserving the high-frequency exposure and opening measures needed for the notification-arrival model.

⁶Home screens, app launchers, and app switchers are observed as foreground apps in the GWS panel, so this rule does not mechanically attribute navigation through these surfaces to notifications.

⁷I begin the main sample in 2021 because Figure 1 shows a sharp level shift in the notification measure around the Android 11 notification user-interface redesign. Since the GWS panel identifies latent push supply through observed notification-induced openings and calibrated click-through rates, a change in notification presentation can alter the measurement relationship between push arrivals and openings. I therefore treat the pre-2021 period as a different measurement regime and use it only to document this break.

Table 1: Notification Model Estimation Samples

App	Panel A. User level				Panel B. User-day level	
	Unique users	App-days per user	Avg. age	Female (%)	Exposure (hours)	Opens (#/day)
Social-media apps (25 apps)						
Messenger	131,108	189.95	39.62	61.61	5.52	0.46
Facebook	122,382	208.75	40.47	61.77	5.41	0.36
YouTube	104,482	224.89	37.65	51.78	5.46	0.11
Instagram	55,155	243.90	35.49	53.95	5.86	0.21
TikTok	50,866	227.69	35.22	58.75	5.70	0.15
Snapchat	49,408	212.31	32.83	56.42	5.95	0.44
<i>Other apps (19 apps)</i>	71,585	417.71	36.64	44.92	5.68	0.21
Communication apps (37 apps)						
Phone	130,590	194.06	40.38	56.91	5.46	0.53
Call	108,819	199.92	40.50	56.98	5.28	1.33
Gmail	96,721	226.82	39.02	57.72	5.64	0.12
Messaging	83,932	215.51	40.15	57.46	5.38	1.08
<i>Other apps (33 apps)</i>	173,382	532.76	40.40	57.33	5.62	0.33

Notes: The table summarizes the app-specific samples used to estimate the notification-arrival model, restricted to user-app-day observations from January 1, 2021 through June 27, 2024. Panel A reports moments after collapsing to unique users within each row; Panel B reports averages at the user-app-day level. Each section lists the largest apps in the category and pools remaining retained apps into “Other.” App-days per user equals the number of user-app-day observations divided by unique users. Exposure is measured in hours per user-app-day. Opens are imputed notification-induced openings per user-app-day. In pooled “Other” rows, users are deduplicated across apps before computing unique users, average age, and female share.

Poisson arrival model. For each app a , I estimate the user-day Poisson model

$$y_{iad} \sim \text{Poisson}\left(p_{g(i)a}\lambda_{iad}T_{iad}\right), \quad \log \lambda_{iad} = \delta_{ia} + \beta_{ia}\tau_d + \mu_{ad}, \quad (10)$$

where y_{iad} is the imputed notification-induced opening count, T_{iad} is exposure time, and λ_{iad} is the latent push-notification arrival rate per unit of exposure. The term $p_{g(i)a}$ is an externally calibrated effective click-through rate: the probability that a push notification for app a delivered to a user in group $g(i)$ generates an opening that the GWS imputation rule records as notification-induced. This object combines behavioral click propensity with systematic proportional bias in the imputation rule.

The latent push-arrival rate has three components. The individual-app intercept δ_{ia} captures persistent cross-user heterogeneity in push supply within an app. This is the main personalized-notification margin. The individual-app slope β_{ia} allows this personalized component to drift with a centered calendar-time index τ_d , accommodating slow changes in user habits or targeting intensity. The app-day fixed effect μ_{ad} absorbs app-wide daily intensity, including platform-wide shocks, seasonality, news events, outages, and other common time variation.

I calibrate $p_{g(i)a}$ using the field experiment in [Bao and Imas \(2026\)](#), which records both delivered push notifications and subsequent app openings. Because the experimental sample is finite, I impose two pooling restrictions for the calibration. First, I estimate click-through rates at the level

of functional app groups—for example, social chat, video, and text-feed apps within social media, and call, SMS/messages, and email apps within communication. Second, I pool demographic cells to the gender level. These restrictions affect only the external calibration of click-through rates; the GWS Poisson model remains app-specific and estimates individual–app heterogeneity from the large observational panel. [Appendix B.2](#) provides details.

Equation (10) is estimated by maximum likelihood with $\log T_{iad}$ and $\log p_{g(i)a}$ entering as offsets. Identification comes from distinct sources of variation. The app–day fixed effect μ_{ad} is identified by common movements in opening rates across users of the same app on the same day. The individual–app intercept δ_{ia} is identified by a user’s average opening rate relative to other users of the same app on the same dates, after normalizing for exposure and calibrated click-through rates. The individual–app slope β_{ia} is identified by systematic changes in that user’s relative opening rate over calendar time, net of app-wide daily shocks. The calibrated click-through rate is not identified from the GWS panel; it fixes the level of latent push supply by converting notification-induced openings into push arrivals.

Daily aggregation. Equation (10) is estimated at the user–app–day level. The daily aggregation is natural for both statistical and economic reasons. Statistically, the Poisson model is closed under aggregation: if notification arrivals are Poisson over disjoint exposure intervals with a common arrival rate conditional on the included covariates, then the daily count is Poisson with mean equal to the arrival rate times total daily exposure. Aggregating within user–app–day therefore preserves the relevant sufficient statistics such as the opening count y_{iad} and exposure time T_{iad} , while avoiding the extreme sparsity of interval-level cells. This is especially important for long-tail apps, where interval-level counts are dominated by zeros and provide little stable information for estimating individual–app heterogeneity and time trends.

Economically, the daily horizon is also the relevant scale for the notification-arrival process I seek to recover. Platforms may pace, batch, or smooth notifications within a day, and users’ attention and app-checking propensities may fluctuate across short intervals. These high-frequency fluctuations are not the object of the notification-supply model. Instead, I aim to recover persistent user-level differences in notification exposure and app-wide daily intensity. A daily panel is the finest level of aggregation that preserves this persistent variation while averaging over within-day timing noise.

4.2 Results

I summarize the estimates in two steps. First, I document the level and composition of the estimated notification environment. Second, I summarize the personalized component, focusing on persistent cross-user dispersion and its relationship with age and gender. [Appendix B](#) reports the full set of notification-estimation outputs.

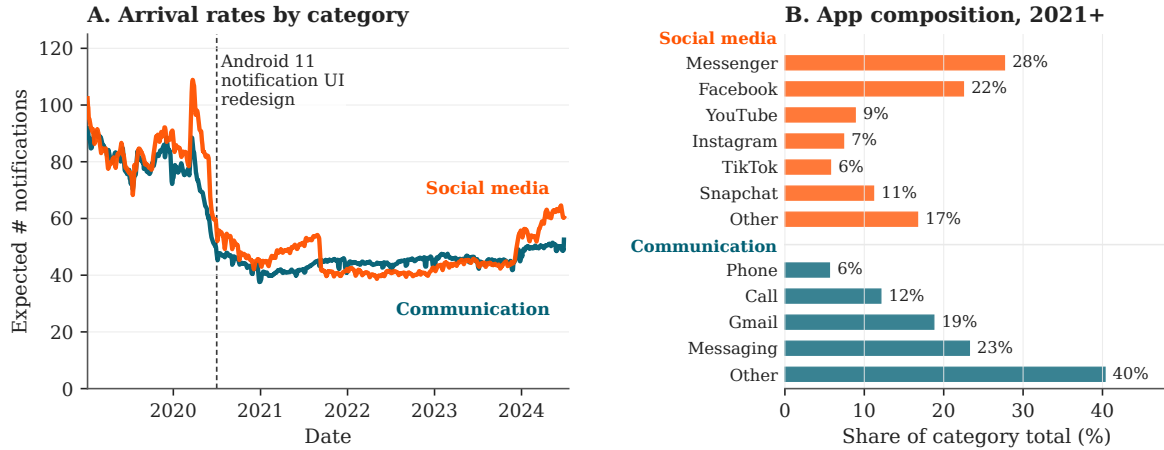


Figure 1: Estimated daily push-notification arrivals

Notes: Panel A reports seven-day centered moving averages of estimated latent push arrivals per retained user-day, summed over retained apps within each category. The vertical dashed line marks the Android 11 notification user-interface redesign. Panel B reports each named app and pooled “Other” row in Table 1 as a share of its category total, computed from the time-series mean of daily category arrivals over January 1, 2021 through June 27, 2024.

Level of notification arrivals. Figure 1 shows that the estimated notification environment is large. Panel A plots daily push-arrival rates for social-media and communication apps. During the post-2021 estimation period, both categories typically generate between 40 and 60 push notifications per retained user-day. The daily category-total mean is 46.0 social-media arrivals and 45.0 communication arrivals; the corresponding medians over dates are 43.9 and 46.8. Communication apps exhibit relatively stable arrival rates over time, while social-media apps display larger fluctuations. The figure also documents the measurement break around the Android 11 notification user-interface redesign, which motivates the post-2021 main sample.

Panel B shows that social-media arrivals are spread across several large apps. Messenger accounts for 27.7 percent of the social-media total, Facebook for 22.5 percent, Snapchat for 11.2 percent, YouTube for 8.9 percent, Instagram for 7.4 percent, and TikTok for 5.7 percent. Communication arrivals are more concentrated in messaging services and residual utilities: Other apps account for 40.3 percent of the communication total, followed by Messaging at 23.2 percent, Gmail at 18.8 percent, Call at 12.1 percent, and Phone at 5.6 percent.

Personalized notifications. The individual-app intercepts and time trends summarize persistent cross-user dispersion in push-arrival rates after netting out app-wide daily intensity. Panel A of Figure 2 reports this dispersion as a mean-normalized personalized push multiplier. Normalizing by the app mean removes level differences across platforms and makes the within-app distributions comparable. The distributions are strongly right-skewed for every app: most users receive below-average personalized push intensity, while a smaller group receives substantially higher arrival rates. Personalized notification supply is therefore not only an app-level phenomenon; it also varies sharply across users within the same app.

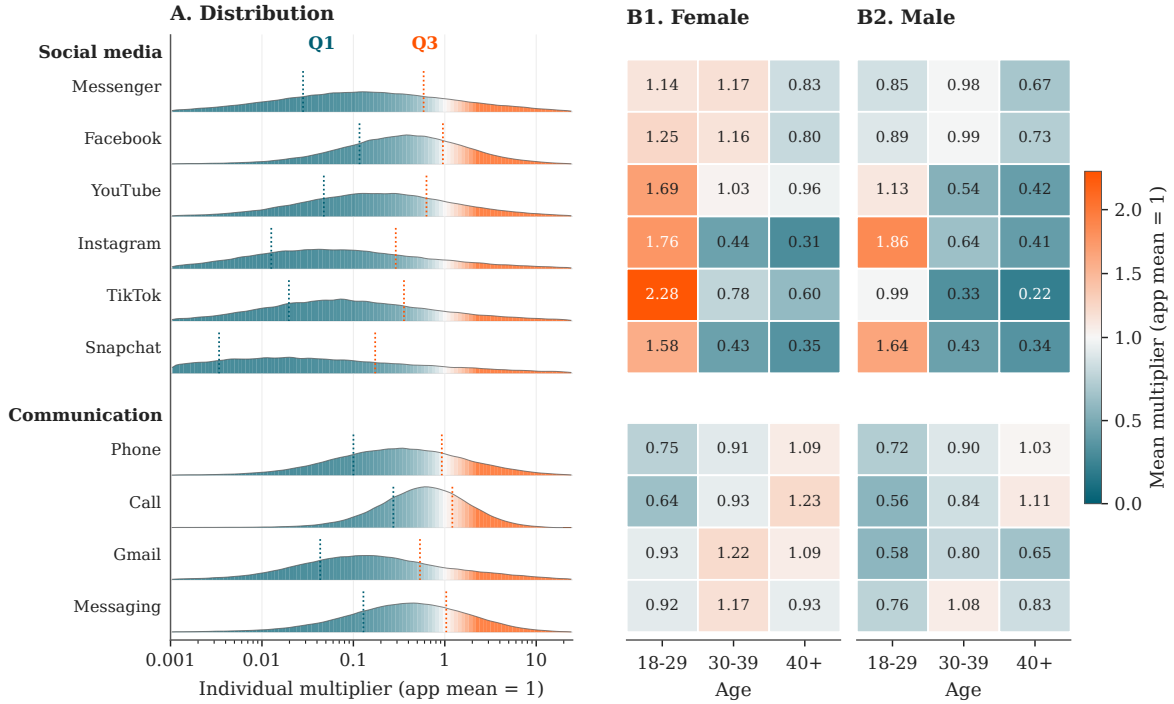


Figure 2: Personalized component: push multipliers and age-gender correlates

Notes: Panel A plots one ridgeline density per app for the mean-normalized personalized push multiplier. For each user-app cell, the multiplier is the 2021-and-later average of $\exp(\delta_{ia} + \beta_{ia}\tau_d)$, divided by the app-specific mean. Ridge height indicates smoothed density on a common horizontal axis. Dotted teal and orange segments mark the first and third quartiles within each app. Panel B reports mean app-normalized multipliers by age group and gender. Orange cells indicate above-average multipliers within app, and teal cells indicate below-average multipliers. Apps are grouped by category and follow the order in Table 1.

Panel B shows that this personalized component is systematically related to demographics. The first pattern is a strong age gradient in social media. For every named social-media app, users aged 18–29 receive higher personalized push intensity than users aged 40 and above. Averaging across gender and named social-media apps, the multiplier is 1.42 for users aged 18–29 and 0.55 for users aged 40 and above. The gradient is especially steep for Instagram, TikTok, Snapchat, and YouTube, and more muted for Facebook Messenger and Facebook.

The second pattern is the reverse age profile for communication apps. Older and middle-aged users receive more communication notifications than younger users. Averaging across gender and named communication apps, the multiplier is 0.73 for users aged 18–29 and 1.00 for users aged 40 and above, with users aged 30–39 also above young users in most communication apps. This contrast suggests that notification personalization is not simply a generic age gradient: younger users are disproportionately targeted or engaged through social-media apps, while older users receive relatively more communication-oriented notifications.

The third pattern is a gender gradient. Women receive higher personalized push multipliers than men in most matched app-age cells, and the difference is particularly consistent in communication apps. Averaging across matched app-age cells, women’s multipliers are roughly 20 percent

higher than men’s.

These demographic gradients are central for the counterfactuals in Section 7. A ban on personalized notification compresses these multipliers toward app-level means, thereby changing not only the level of notification supply but also its allocation across age and gender groups.

5 Production: The Costs of Interruptions

The dynamic cost of interruptions identified by the framework in Section 3 operates through the production-duration function $\Delta(X, m, L)$: each additional interruption lengthens the current delivery trip, raises cumulative spell time, lowers the worker’s continuation probability, and reduces the expected number of trips completed during the spell. This section estimates that production margin in the setting of food-delivery couriers. The setting is useful because output is organized into repeated trips, trip distance and duration are directly measurable from GPS traces, and workers perform the task through the same smartphone environment that generates non-work interruptions. The estimates below quantify how phone interruptions affect delivery duration, providing the production input for the structural model and the counterfactual policies.

5.1 Food-delivery courier sample

Food-delivery couriership is a central setting in app-mediated work. The broader U.S. couriers-and-messengers sector employs roughly one million payroll workers, while platform-based delivery adds a large non-payroll participation margin: DoorDash reports that more than seven million U.S. Dashers earned over \$16 billion on the platform in 2024, with the average Dasher spending less than four hours per week on delivery.⁸ The sector is therefore economically important, but it is also empirically attractive for studying attention and productivity. Delivery work consists of repeated, time-stamped tasks; the worker’s route, distance, and completion time are observable; and the production technology requires frequent interaction with delivery and navigation apps, creating a natural environment in which non-work phone activity can interfere with output.

I construct a sample of food-delivery couriers from the GWS panel. Couriers are identified through foreground use of the driver-facing apps “Dasher” and “Grubhub for Drivers.”⁹ For each courier, I impute delivery trips from contiguous GPS movement bouts anchored to delivery-app foreground episodes. I then connect adjacent trips separated by gaps of at most 60 minutes into a working spell, producing the alternating sequence of delivery trips and idle periods used in the framework of Section 3.¹⁰ The resulting regression sample contains 9,270 couriers, 345,414 working spells, and 848,159 delivery trips. In those identified trips and spells, I define work-related

⁸Sources: U.S. Bureau of Labor Statistics, *Industries at a Glance: Couriers and Messengers, NAICS 492*, Current Employment Statistics; and DoorDash, *Measuring Our Impact: Unveiling the 2024 Economic Impact Report*.

⁹I exclude the “Uber Driver” app because it combines food delivery with ride-hailing activity, making it difficult to isolate food-delivery production.

¹⁰[Appendix C.1](#) documents the trip and spell construction in detail.

Table 2: Summary Statistics for Food-Delivery Courier Sample

Variable	Obs.	Mean	Std.	Min	Median	Max
Panel A. Worker (N=9,270)						
Age	9,070	37.05	10.12	18.00	36.00	93.00
Female (%)	8,989	57.88	49.38	0.00	100.00	100.00
Number of spells	9,270	37.26	74.10	1.00	13.00	1501.00
Panel B. Spell (N=345,414)						
Number of trips	345,414	2.46	1.94	1.00	2.00	22.00
Duration (min)	345,414	88.42	89.29	3.00	60.47	642.42
Miles	345,414	11.76	12.64	0.15	7.39	102.39
Screen time: work (min)	345,414	67.50	70.37	0.50	43.98	605.12
Screen time: non-work (min)	345,414	12.86	19.35	0.00	4.83	286.45
Social media	345,414	3.22	8.80	0.00	0.00	203.12
Communication	345,414	2.67	5.43	0.00	0.60	133.03
Interruptions: non-work	345,414	8.89	12.01	0.00	4.00	327.00
Social media	345,414	1.76	4.66	0.00	0.00	150.00
Communication	345,414	3.26	5.54	0.00	1.00	204.00
Panel C. Delivery trip (N=848,159)						
Duration (min)	848,159	24.94	25.92	3.00	15.55	231.20
Miles	848,159	4.79	5.38	0.15	2.90	88.06
Screen time: work (min)	848,159	21.26	22.96	0.40	12.87	202.05
Screen time: non-work (min)	848,159	2.30	3.49	0.00	1.20	61.02
Social media	848,159	0.47	1.52	0.00	0.00	31.25
Communication	848,159	0.58	1.42	0.00	0.00	31.30
Interruptions: non-work	848,159	2.16	2.77	0.00	1.00	24.00
Social media	848,159	0.37	1.00	0.00	0.00	22.00
Communication	848,159	0.84	1.61	0.00	0.00	23.00

Notes: The table reports summary statistics for the food-delivery courier sample used in the production-function regressions after complete-case filtering and fixed-effect singleton pruning. Panel A is at the worker level, Panel B is at the working-spell level, and Panel C is at the delivery-trip level. Panel B summarizes full working spells represented by retained regression-sample trips. The set of work apps includes “Dasher”, “Grubhub for Drivers”, “Google Maps”, “Waze”, and “Camera”.

apps as “Dasher,” “Grubhub for Drivers,” “Google Maps,” “Waze,” and “Camera.”¹¹ An interruption is a transition into a non-work app that either initiates a screen session or interrupts an ongoing work-app episode. I measure screen time and interruptions at the app level and aggregate them into app categories. Trip distance is measured from GPS traces.

Table 2 summarizes the sample. Panel A reports worker-level characteristics. The average courier is 37.05 years old, and 57.88 percent of couriers are female. These moments are broadly comparable to company-reported Dasher demographics: DoorDash reports that 55 percent of Dashers are women and that the median age of Dashers is between 35 and 44.¹² The median courier appears in 13 working spells, while the mean is 37.26 spells, reflecting the highly skewed participation patterns typical of app-mediated work.

Panel B describes working spells. The average spell lasts 88.42 minutes, contains 2.46 delivery

¹¹The camera is included because couriers often use it to document drop-off completion.

¹²DoorDash, *Measuring Our Impact: Unveiling the 2024 Economic Impact Report*, p. 25. The age distribution is based on Dasher accounts that delivered an order through DoorDash in 2024.

trips, and covers 11.76 miles. The median spell is shorter, at 60.47 minutes and two trips, indicating substantial right-tail variation in spell length. Phone activity is observed for a large share of spell time: work-app and non-work-app foreground episodes together account for 80.36 minutes, or 90.9 percent of the average spell. This dense measurement is important for the empirical design because it allows me to distinguish activities related to production from activities related to non-work apps during the same working spell. Non-work phone use is prevalent. During the average spell, couriers spend 12.86 minutes on non-work apps, including 3.22 minutes on social-media apps and 2.67 minutes on communication apps. The average spell contains 8.89 non-work interruptions, or about 1.01 interruptions per 10 minutes of spell time. Social-media apps account for 19.80% interruptions per spell, while communication apps account for 36.67%. These moments show that non-work phone activity is a regular feature of delivery work, not an exceptional event concentrated in a small number of spells.

Panel C reports the corresponding delivery-trip statistics. The average trip lasts 24.94 minutes and covers 4.79 miles, with a median duration of 15.55 minutes and median distance of 2.90 miles. App activity is even more densely measured during trips than during spells: work-app and non-work-app foreground episodes together account for 23.56 minutes, or 94.5 percent of average trip duration. During the average delivery trip, couriers spend 21.26 minutes (85.24%) on work apps and 2.30 minutes (9.22%) on non-work apps. They experience 2.16 non-work interruptions, corresponding to 0.87 interruptions per 10 trip minutes, below the spell-level rate. The lower interruption rate during delivery trips is consistent with the model’s central trade-off: couriers continue to derive utility from phone activity while working, but the opportunity cost of non-work phone use is higher during production than during idle time. These trip-level moments show that non-work phone activity is not confined to downtime; it occurs during the delivery trips used to estimate production-duration function.

5.2 Estimation strategy

I estimate the trip-duration function using the trip-level panel described above. The empirical goal is to recover the conditional mapping from interruptions, trip distance, and cumulative spell time into delivery duration for the structural model, where the coefficient on interruptions is the key margin for policy interpretation. The fitted production function maps directly into $\Delta_{is}(X, m, L)$: interruptions are costly because they lengthen the current delivery trip, raise cumulative spell time at subsequent decisions, and lower the probability that the worker continues the spell. The worker–spell subscript absorbs production shifters that are held fixed in the structural implementation, including worker heterogeneity, local time conditions, weather, and pay incentives.

Specification. For each delivery trip t in worker–spell (i, s) , I parameterize per-mile trip duration as

$$\log\left(\frac{\ell_{ist}}{m_{ist}}\right) = \beta^X \left(\frac{X_{ist}}{m_{ist}}\right) + \mathcal{Q}(m_{ist}; \beta^m) + \mathcal{Q}(L_{ist}; \beta^L) + \mathbf{W}'_{ist} \beta^W + \mathcal{F}_{ist} + \varepsilon_{ist}, \quad (11)$$

where ℓ_{ist} is trip duration, m_{ist} is trip distance, X_{ist} is the number of non-work interruptions during the trip, and L_{ist} is cumulative spell duration at the start of the trip. The dependent variable is log minutes per mile, and the main phone-use regressor is interruptions per mile, X_{ist}/m_{ist} . The function $\mathcal{Q}(z; \beta)$ denotes a quadratic in z , so the specification flexibly controls for nonlinear effects of trip distance and cumulative spell duration. The vector $\mathbf{W}_{ist}$ includes weather controls, measured by daily average temperature and precipitation, as well as payoff per mile. The fixed-effect component $\mathcal{F}_{ist}$ includes worker, trip-date, and trip-start-hour fixed effects. Identification therefore comes from within-worker variation across trips after absorbing common daily shocks and systematic hour-of-day differences.

The coefficient β^X is the semi-elasticity of trip duration per mile with respect to interruptions per mile. Thus, for a trip of length m , one additional interruption changes log trip duration by β^X/m , holding distance, cumulative spell time, controls, and fixed effects fixed. A positive β^X implies that interruptions lower production efficiency by increasing minutes per mile. The distance terms allow per-mile duration to vary flexibly with route length, while the cumulative-duration terms capture systematic changes in delivery productivity over the course of a working spell. In the structural model, the fitted version of equation (11) provides the empirical counterpart to $\Delta_{is}(X, m, L)$, with the remaining controls and fixed effects summarized by the worker–spell-specific production shifter. Section ?? describes this mapping in detail.

Identification. A central identification concern is that interruptions are endogenous to trip conditions. Slow trips may create more opportunities for phone use: traffic, restaurant delays, or parking frictions can lengthen a trip and also induce the courier to check the phone. In that case, OLS would overstate the productivity cost of interruptions. Conversely, workers may avoid non-work phone use during difficult trips, which could attenuate the OLS relationship. I therefore instrument interruptions per mile using two sources of predetermined variation: an app-level shift-share interruption shock and the worker’s lagged in-trip interruption rate.

The first instrument captures app-side shocks to the attractiveness or salience of non-work apps. Let $X_{a,c,h}^{-i}$ denote the leave-one-out average number of interruptions into app a among GWS panelists other than worker i in state c and calendar-hour cell h . These aggregate app-time shocks capture variation such as outages, viral content, news events, or platform-wide engagement shocks. The leave-one-out construction ensures that worker i ’s own phone use does not mechanically enter the shock. I then interact these app-level shocks with predetermined worker-specific exposure shares. Let $s_{ia,t-}$ be the share of worker i ’s non-work interruptions accounted for by app a during the two weeks preceding trip t .¹³ The shift-share instrument is

$$Z_{ist}^{SS} = \sum_{a \in \mathcal{A}} s_{ia,t-} X_{a,c(i),h(ist)}^{-i}, \quad (12)$$

where $\mathcal{A}$ is the set of top 50 non-work apps, $c(i)$ is worker i ’s state, and $h(ist)$ is the calendar-hour

¹³The two-week window captures stable app habits while excluding current-trip phone use. The estimates are robust to alternative pre-period windows.

cell corresponding to trip t .¹⁴ The instrument is high when apps that are important in worker i 's pre-period phone portfolio experience unusually high interruption activity among other users in the same state and hour. [Appendix C.2](#) illustrates this variation using the major Facebook outage on October 4, 2021: when leave-one-out Facebook interruption activity falls sharply, the shift-share instrument falls most for couriers with high pre-period Facebook exposure, and the first stage shows a corresponding decline in their in-trip interruptions.¹⁵

The second excluded instrument is the worker's lagged in-trip interruption rate,

$$Z_{ist}^{Lag} = \frac{1}{|\mathcal{T}_{i,t-}|} \sum_{\tau \in \mathcal{T}_{i,t-}} \frac{X_{i\tau}}{m_{i\tau}}, \quad (13)$$

where $\mathcal{T}_{i,t-}$ is the set of worker i 's delivery trips in the two weeks preceding trip t . This instrument captures persistent individual exposure to phone interruptions that is predetermined relative to the current trip. It is relevant because phone habits and adaptive notification supply make recent interruption behavior predictive of current interruptions, as documented in [Section 4.2](#).

The first stage uses Z_{ist}^{SS} and Z_{ist}^{Lag} as excluded instruments for X_{ist} / m_{ist} , together with the same controls and fixed effects as in equation (11). The exclusion restriction is that, conditional on worker, date, and start-hour fixed effects and the observed trip controls, these instruments affect trip duration only through current in-trip interruptions. For the shift-share instrument, the identifying assumption is shock exogeneity: app-level shocks among other panelists are unrelated to worker i 's unobserved trip productivity shocks after conditioning on the controls.¹⁶ The leave-one-out aggregate averages over idiosyncratic user-level shocks, while predetermined app shares allocate common app-side shocks across couriers according to pre-existing exposure rather than current working conditions. For the lagged-rate instrument, the maintained restriction is that pre-period interruption rates affect current trip duration only through current interruptions, after controlling for worker fixed effects and observed trip conditions.

5.3 Results

Table 3 reports estimates of equation (11). The main result is that phone interruptions substantially slow delivery production. Column (1) reports the OLS benchmark, while column (2) reports the baseline IV specification. To interpret magnitudes consistently, I translate the estimates into the effect of one additional interruption on the average trip, which is 4.8 miles and lasts 24.9 minutes. In the OLS, one additional interruption raises trip duration by 3.8 percent, or about 0.9 minutes. The baseline IV estimate is larger: one additional interruption raises trip duration by 4.8 percent,

¹⁴For feasibility, I construct the shift-share instrument using the top 50 non-work apps ranked by total screen time. The estimates are robust to alternative choices of the app set.

¹⁵Although the outage events provide attractive and interpretable shocks to interruption exposure, these events are short-term and rare, and therefore lack of statistical power in making inference.

¹⁶This design follows the shock-exogeneity logic of shift-share research designs in [Borusyak, Hull, and Jaravel \(2022, 2025\)](#); [Adao et al. \(2019\)](#) discuss the inference issues that arise when many units are exposed to common shocks. [Appendix C.2](#) reports diagnostics for the instrument vector, including within-worker dispersion, app-share concentration, and residualized first-stage relationships.

Table 3: Production Consequences of Phone Interruptions

	(1) Duration per mile (log) OLS	(2) Duration per mile (log) IV	(3) Duration per mile (log) IV	(4) Stop time per mile (min) IV	(5) Avg. speed (mph) IV
Endogenous variables					
Interruptions per mile (#)	0.177 (0.003)	0.226 (0.006)	0.311 (0.022)	4.168 (0.063)	7.096 (0.179)
Screen time per mile (min)			-0.087 (0.020)		
Trip-level controls					
Miles (mi)	-0.074 (0.001)	-0.069 (0.001)	-0.073 (0.001)	0.271 (0.008)	1.982 (0.043)
Miles square (mi ²)	0.002 (0.000)	0.002 (0.000)	0.002 (0.000)	-0.007 (0.000)	-0.050 (0.002)
Cumulative time (hr)	-0.012 (0.002)	-0.010 (0.002)	0.001 (0.003)	0.211 (0.011)	0.554 (0.046)
Cumulative time square (hr ²)	0.001 (0.000)	0.000 (0.000)	-0.001 (0.000)	-0.031 (0.002)	-0.061 (0.008)
Weather and payoff controls	Yes	Yes	Yes	Yes	Yes
FE: worker, date, and trip-hour	Yes	Yes	Yes	Yes	Yes
Observations	848,159	848,159	848,159	848,159	848,159
Mean dep. var.	1.752	1.752	1.752	1.346	15.587
First-stage F-stat	–	325.65	104.51	325.65	325.65

Notes: The table reports trip-level estimates for food-delivery couriers. Columns (1)–(3) use log trip duration per mile as the dependent variable: column (1) reports the OLS benchmark, column (2) reports the baseline IV specification, and column (3) adds non-work screen time per mile as a second endogenous phone-use measure. Columns (4)–(5) apply the baseline IV specification to stop time per mile and average travel speed. The baseline instrument set includes the shift-share interruption shock and lagged interruptions per mile; column (3) adds analogous instruments based on shift-share screen-time shocks and lagged screen time per mile. All specifications include worker, trip-date, and trip-start-hour fixed effects, as well as weather and payoff controls. The standard deviation of interruptions per mile is 1.193. The first-stage F-statistic reports the excluded-instrument F-statistic in columns (2), (4), and (5), and the clustered joint excluded-instrument Wald statistic in column (3). Standard errors are clustered by worker and reported in parentheses.

or about 1.2 minutes. The first stage is strong, with an excluded-instrument F-statistic of 325.7.

The IV estimate implies economically meaningful production losses. Delivery work consists of repeated trips within a spell, so a 1.2-minute delay on the current trip also has a dynamic consequence: it raises cumulative spell time and reduces the number of trips the courier can complete before stopping. The fact that the IV estimate exceeds the OLS estimate suggests that the net OLS bias is downward. This pattern is consistent with measurement error in interruptions or with behavioral avoidance where couriers may reduce discretionary phone use during trips that are already difficult.

Column (3) assesses whether the interruption effect is simply proxying for total non-work screen time. I add non-work screen time per mile as a second endogenous phone-use measure and instrument it with the corresponding shift-share and lagged screen-time instruments. The interruption coefficient remains positive and becomes larger: one additional interruption on the average trip raises duration by 6.7 percent, or about 1.7 minutes. The screen-time coefficient is

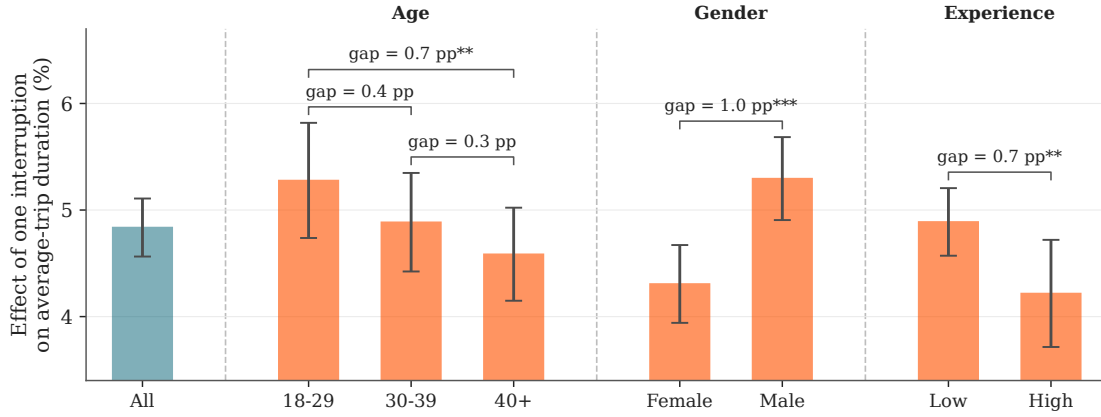


Figure 3: Heterogeneity in the production cost of interruptions

Notes: Bars report the implied percent change in average-trip duration from one additional interruption, computed as $100 \times [\exp(\hat{\beta}/4.79) - 1]$, where 4.79 is the average trip distance in miles and $\hat{\beta}$ is the interruption coefficient from equation (11). The teal bar uses the pooled estimate from column (2) of Table 3; orange bars use group-specific coefficients from fully interacted IV versions of equation (11). The dependent variable is log trip duration per mile, and the endogenous regressor is non-work interruptions per mile. Each interacted specification includes the controls and fixed effects from Table 3. Whiskers report the 95 percent confidence interval, with standard errors clustered by worker. Stars indicate significance from pairwise coefficient-equality tests at the 10, 5, and 1 percent levels.

negative. I therefore interpret column (3) as a robustness exercise rather than as a second structural primitive. The result indicates that the production cost is tied to discrete attention shifts into non-work apps, not merely to the total duration of non-work screen exposure.

Columns (4)–(5) examine the channels through which interruptions affect trip duration. Column (4) replaces the dependent variable with stop time per mile, constructed from GPS movement traces.¹⁷ Translating the estimate to the same average-trip metric, one additional interruption raises stop time by 0.87 minutes per mile, or about 4.2 minutes over the average trip. Column (5) shows that couriers partly compensate by moving faster: one additional interruption on the average trip raises average travel speed by 1.5 miles per hour. These estimates point to a catch-up pattern. Interruptions increase stationary time, and couriers partially offset this lost time by moving faster during the travel portion of the trip. The stationary-delay channel dominates this catch-up response, generating the net increase in total trip duration reported in column (2).¹⁸

Heterogeneity. Figure 3 reports heterogeneity in the production cost of interruptions. I estimate fully interacted IV versions of equation (11) by age group, gender, and courier experience, where high-experience couriers are those above the 90th percentile of average weekly trip miles. For comparability with the main estimates, I again interpret each coefficient as the effect of one additional interruption on the average trip.

¹⁷Stop time is the amount of trip time classified as stationary from GPS traces, divided by trip miles. [Appendix C.1](#) describes the construction.

¹⁸The speed response may also have safety implications. I do not observe crashes, vehicle mode, posted speed limits, or road type, so I do not monetize this channel in the main welfare analysis. Incorporating safety externalities would strengthen, rather than weaken, the case for policies that reduce in-trip interruptions.

The heterogeneity estimates reveal three patterns. First, the age gradient is present. One additional interruption raises duration by 5.3 percent for couriers aged 18–29, compared with 4.6 percent for couriers aged 40 or older; the difference is statistically significant at the 5 percent level. Second, the gender gradient is larger. The same interruption raises duration by 5.3 percent for men and 4.3 percent for women, with the difference significant at the 1 percent level. Third, experience attenuates the cost of interruptions. One additional interruption raises duration by 4.9 percent for couriers below the high-experience threshold, compared with 4.2 percent for high-experience couriers; this difference is significant at the 5 percent level.

These patterns imply that the productivity burden of phone interruptions is unevenly distributed across workers. Younger and male couriers face larger duration penalties from a given interruption, and Section 4.2 shows that these same groups also receive higher social-media notification rates. The exposure and production margins therefore reinforce each other: groups that receive more notifications are also groups for whom interruptions are more costly. Experience provides a separate source of heterogeneity, suggesting that more experienced couriers are better able to manage interruptions or absorb their production consequences. This heterogeneity matters for the policy discussion in Section 7: supply-side notification policies affect not only the average number of interruptions, but also their allocation across workers with different production costs.

6 Preferences for Interruptions

The previous two sections recover the measured primitives of the worker’s interruption problem. Section 4 estimates category-specific notification arrival rates, which determine the supply of external prompts in a spell. Section 5 estimates the production-duration function, $\hat{\Delta}_{is}(X, m, L)$, which maps interruptions, trip distance, and cumulative spell time into delivery duration. This section completes the structural model by estimating the remaining behavioral objects: the flow utility from phone activity, the continuation probability, and the doer’s revealed-preference value of output.

Throughout the section, $\mathcal{M} = \{SM, CM, OT\}$ indexes social-media, communication, and other apps, while $\mathcal{M}^* = \{SM, CM\}$ indexes the app categories with external notification arrivals. I estimate the model separately for each demographic group

$$g \in \mathcal{G} = \{\text{female, male}\} \times \{18\text{--}29, 30\text{--}39, 40+\},$$

allowing all preference parameters to vary by group. To keep notation readable, I omit the group subscript g whenever it is unambiguous.

The section proceeds in four steps. Section 6.1 converts the notification and production estimates into model-ready inputs and documents how these inputs vary across demographic groups. Section 6.2 specifies the finite action set, flow utility, continuation probability, and payoff shocks used in the likelihood. Section 6.3 describes estimation and identification. Section 6.4 reports the estimated preference parameters and interprets the main behavioral patterns.

Table 4: Structural-Model Sample and Inputs by Demographic Group

Variable	Female			Male		
	18–29	30–39	40+	18–29	30–39	40+
Panel A. Structural sample						
Workers	737	1,494	1,413	620	942	958
Working spells	17,543	40,011	52,576	18,488	35,793	44,859
Idle periods	22,225	51,093	69,843	27,161	52,322	62,661
Delivery trips	39,781	91,124	122,435	45,672	88,132	107,570
Panel B. Working spell						
Duration (min)	86.6	87.2	94.6	87.0	92.3	94.3
Delivery trips per spell	2.27	2.28	2.33	2.47	2.46	2.40
Social-media notification arrivals (#/hr)	4.21	2.41	1.54	3.99	2.12	1.45
Communication notification arrivals (#/hr)	2.02	2.66	2.69	2.08	2.37	2.72
Net payoff per mile (\$/mi)	1.70	1.73	1.71	1.73	1.77	1.77
Panel C. Idle period						
Duration (min)	18.7	18.3	18.9	17.9	18.3	18.4
Social-media interruptions (#)	0.88	0.57	0.39	0.94	0.64	0.36
Communication interruptions (#)	0.67	0.76	0.76	0.57	0.70	0.74
Other-app interruptions (#)	1.16	1.10	1.04	1.16	1.22	1.10
Panel D. Delivery trip						
Duration (min)	27.8	28.0	29.9	24.6	26.6	28.6
Trip distance (mi)	5.38	5.43	5.65	4.80	5.32	5.41
Social-media interruptions (#)	0.61	0.43	0.31	0.72	0.44	0.29
Communication interruptions (#)	0.80	0.95	1.01	0.67	0.83	0.98
Other-app interruptions (#)	1.13	1.13	1.09	1.04	1.13	1.13

Notes: The table summarizes the final structural estimation sample and model inputs by gender–age group. The sample is restricted to 2021-and-later delivery spells that can be matched to notification-arrival rates, production-duration inputs, payoff and weather controls, and observed interruption choices. Panel A reports counts. Panels B–D report means at the working-spell, idle-period, and delivery-trip levels, respectively. Notification arrivals are latent push-arrival rates constructed from the notification model and measured per hour at the spell level. Interruption rows report period-level counts. Net payoff per mile is compensation net of gasoline cost.

6.1 Structural inputs by demographic group

Before estimating preferences, I convert the reduced-form estimates from Sections 4 and 5 into model-ready inputs. These inputs are not preference estimates; they are measured state variables and transition objects that enter the structural likelihood. I estimate preferences separately by demographic group. The two central inputs are the category-specific notification arrival rates, $\{\hat{\lambda}_{is}^{SM}, \hat{\lambda}_{is}^{CM}\}$, and the production-duration function, $\hat{\Delta}_{is}(X, m, L)$. The notification rates determine the distribution of external notification constraints faced by worker i in spell s . The production function maps each candidate interruption choice into trip duration and hence into completed spell time. This subsection describes how these objects are constructed and documents the heterogeneity in the final structural sample.

Table 4 summarizes the final sample used in the structural estimation. The panel is restricted to delivery spells beginning in 2021 or later that can be matched to notification-arrival inputs, production-duration inputs, payoff and weather controls, and observed interruption choices. The

retained sample contains all six gender–age cells, with 6,164 workers and 209,270 working spells in total. Sample size is heterogeneous across groups: women aged 40 or older contribute the largest number of spells, while the youngest male and female groups are smaller but still contain more than 600 workers each.

The table previews the two sources of heterogeneity that the structural model must separate. First, notification supply differs sharply across demographic groups. Social-media notification arrivals are highest for younger workers and decline steeply with age: workers aged 18–29 receive about four social-media notifications per hour, compared with roughly 1.5 per hour among workers aged 40 or older. Communication notification arrivals are flatter and tilt toward older workers. Second, observed phone choices differ between idle periods and delivery trips. Social-media interruptions are more common among younger workers, while communication interruptions are relatively more concentrated among middle-aged and older workers. Because delivery trips are longer than idle periods, the period-level counts in Panels C and D imply a lower interruption intensity during production than during idle time. This idle-versus-trip contrast is the central behavioral margin used below to identify the revealed-preference value of output, α .

Notification arrivals. Equation (10) is estimated at the user–app–day level, while the structural model requires worker–spell arrival rates for the notification-generating app categories, social media and communication. I map app–day rates into spell-level category rates in three steps. First, I evaluate the fitted latent push-arrival rate for every retained user–app–day observation from the notification model. Second, I allocate each fitted app–day rate across clock hours using a user–app-specific hourly profile. Third, I average the resulting hourly rates over the exact working-spell interval and aggregate across raw apps within category.

Formally, for category $c \in \{SM, CM\}$,

$$\hat{\lambda}_{is}^c = \frac{1}{|\mathcal{I}_{is}|} \sum_{a \in \mathcal{A}_c} \int_{\mathcal{I}_{is}} \hat{\lambda}_{ia,d(\tau)} \pi_{ia,h(\tau)} d\tau, \quad (14)$$

where $\mathcal{I}_{is}$ is the clock-time interval covered by spell s , $\mathcal{A}_c$ is the set of raw apps in category c , $\hat{\lambda}_{ia,d(\tau)}$ is the fitted latent push-arrival rate for app a on date $d(\tau)$, and $\pi_{ia,h(\tau)}$ is the normalized hourly profile for user i and app a . [Appendix B.3](#) provides implementation details.

Figure 4 shows the distributional version of the notification patterns summarized in Table 4. Social-media notification supply is both high and dispersed among young workers. The mean is 4.21 arrivals per hour for women aged 18–29 and 3.99 for men aged 18–29, compared with 1.54 and 1.45 arrivals per hour for women and men aged 40 or older. Communication notification supply has a different demographic profile: means range from 2.02 to 2.72 arrivals per hour and are highest for older workers. The distributions are right-skewed in every group, so the mean exceeds the median, especially for young workers’ social-media notifications.

These differences matter because social-media and communication notifications shift different external constraints in the worker’s action set. A worker with a higher $\hat{\lambda}_{is}^{SM}$ faces more opportu-

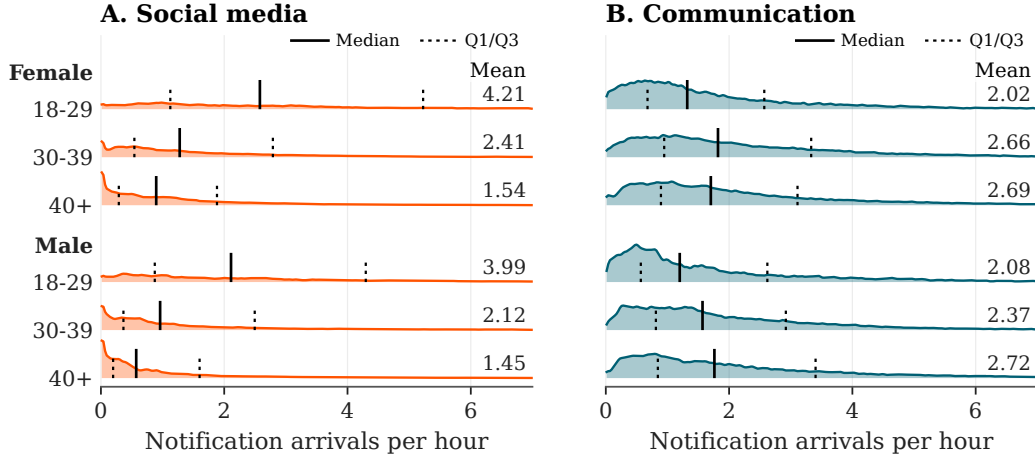


Figure 4: Notification-arrival distributions by app category and demographic group

Notes: The figure reports worker–spell distributions of model-input notification-arrival rates, measured in arrivals per hour. The structural period panel is collapsed to one row per worker–spell, and each worker–spell receives equal weight. Rows report the six gender–age groups used in the structural estimation. Solid vertical segments mark medians; dashed vertical segments mark first and third quartiles. Row labels on the right report group means. Densities are drawn over 0 to 7 arrivals per hour; means and quantiles are computed using the full distribution.

nities for externally triggered social-media interruptions; a worker with a higher $\hat{\lambda}_{is}^{CM}$ faces more opportunities for externally triggered communication interruptions. The structural model uses this variation to distinguish preferences over interruptions from differences in the opportunities to be interrupted.

Production-duration function. The second input is the production-duration function. I use the baseline IV specification in column (2) of Table 3 and re-estimate it separately by demographic group. This yields group-specific production parameters, including the interruption coefficient, the distance profile, and the cumulative-spell-time profile. For a candidate action $\mathbf{x}$ during a trip with distance m and cumulative spell time L , the fitted production function maps the implied total number of interruptions, $X(\mathbf{x})$, into predicted trip duration:

$$\hat{\ell}_{g,is}(\mathbf{x}; m, L) = m \hat{\Delta}_{g,is}(X(\mathbf{x}), m, L). \quad (15)$$

The worker–spell subscript absorbs production shifters that are fixed when the worker chooses phone activity, including worker heterogeneity, time effects, weather, payoff conditions, and other controls from the production regression. Table 5 reports the group-specific production parameters that enter this mapping.

Table 5 shows that interruptions lengthen delivery trips in every demographic group, but the magnitude varies in economically meaningful ways. Translating the interruption coefficient into the effect of one additional interruption on the pooled average trip, the implied duration increase ranges from 3.88 percent, or 0.97 minutes, for women aged 40 or older to 5.54 percent, or 1.38 minutes, for men aged 18–29. Among women, the production penalty declines with age, from

Table 5: Production-Duration Inputs by Demographic Group

	Female			Male		
	18–29	30–39	40+	18–29	30–39	40+
<i>Dependent variable: log trip duration per mile</i>						
Interruptions per mile, X/m (#)	0.235 (0.020)	0.220 (0.012)	0.182 (0.013)	0.258 (0.016)	0.255 (0.015)	0.241 (0.013)
Trip miles, m (mi)	–0.075 (0.003)	–0.069 (0.003)	–0.068 (0.002)	–0.073 (0.003)	–0.069 (0.002)	–0.068 (0.003)
Trip miles squared, m^2 (mi ²)	0.002 (0.000)	0.002 (0.000)	0.002 (0.000)	0.002 (0.000)	0.002 (0.000)	0.002 (0.000)
Cumulative time, L (hr)	–0.001 (0.005)	–0.013 (0.004)	–0.012 (0.004)	–0.007 (0.005)	–0.013 (0.004)	–0.007 (0.004)
Cumulative time squared, L^2 (hr ²)	–0.002 (0.001)	0.001 (0.001)	0.001 (0.001)	0.000 (0.001)	0.001 (0.001)	0.000 (0.001)
Implied effect of one interruption (%)	5.04	4.71	3.88	5.54	5.46	5.17
Implied effect of one interruption (min)	1.26	1.17	0.97	1.38	1.36	1.29

Notes: The table reports group-specific production-duration inputs used in the structural model. Each column estimates the baseline IV specification in equation (11) separately for the indicated gender–age group. The dependent variable is log trip duration per mile, and the endogenous regressor is interruptions per mile. The excluded instruments are the shift-share interruption shock and lagged interruptions per mile. All specifications include weather controls, payoff per mile, and worker, trip-date, and trip-start-hour fixed effects, which are absorbed into a spell-specific shifter into the structural model. Standard errors are clustered by worker and reported in parentheses. The implied-effect rows translate the interruption coefficient into the effect of one additional interruption on the pooled average trip, using an average trip duration of 24.94 minutes. These rows are derived magnitudes, not additional regression coefficients.

5.04 percent for ages 18–29 to 3.88 percent for ages 40 or older. Among men, the penalty is higher and more stable across age groups, ranging from 5.17 to 5.54 percent.

The other production coefficients govern how the cost of interruptions is translated into trip duration over the state space. In particular, the distance coefficients should be read on the per-mile productivity scale: because the dependent variable is log duration per mile, the negative trip-mile coefficients imply that longer trips have lower average minutes per mile, consistent with fixed pickup, parking, and drop-off components being spread over more miles. They do not imply that longer trips take less total time.

Taken together, the notification and production inputs imply that exposure and productivity costs vary along related demographic dimensions. Younger workers face higher social-media notification arrival rates, while male workers face larger production penalties from a given interruption. The structural model uses these inputs to separate three distinct sources of heterogeneity: opportunities to be interrupted, the production cost of interruptions, and preferences over interruptions.

6.2 Structural parameterization

I now specify the finite action set, flow utility, continuation probability, and payoff shocks used in the structural likelihood. The model is estimated separately by demographic group $g \in \mathcal{G}$, but I

suppress the group subscript in this subsection to simplify notation.

Action set. In each period, the worker chooses the interruption vector, $\mathbf{x}_t = (\mathbf{x}_t^{ex}, \mathbf{x}_t^{in})$. The external component, $\mathbf{x}_t^{ex} = (x_{ct}^{ex})_{c \in \mathcal{M}^*}$, records how many incoming notifications the worker acts on in the social-media and communication categories. The internal component, $\mathbf{x}_t^{in} = (x_{ct}^{in})_{c \in \mathcal{M}}$, records self-initiated checks in all three app categories. For computation, I cap each external notification state at $\bar{n}$ and each internal check count at $\bar{x}_c^{in}$. Given notification vector $\mathbf{n}_t$, the finite feasible set is

$$\mathcal{A}(\mathbf{n}_t) = \left\{ \mathbf{x}_t = (\mathbf{x}_t^{ex}, \mathbf{x}_t^{in}) : x_{ct}^{ex} \in \{0, \dots, n_{ct}\} \forall c \in \mathcal{M}^*, \quad x_{ct}^{in} \in \{0, \dots, \bar{x}_c^{in}\} \forall c \in \mathcal{M} \right\}. \quad (16)$$

The caps are computational top-codes set high enough that only a thin tail of the empirical distribution is censored. Recall the aggregation of interruptions:

$$x_{ct}(\mathbf{x}_t) = x_{ct}^{in} + \mathbb{1}\{c \in \mathcal{M}^*\} x_{ct}^{ex} \quad \text{and} \quad X(\mathbf{x}_t) = \sum_{c \in \mathcal{M}} x_{ct}(\mathbf{x}_t).$$

Flow utility. Flow utility from phone activity is a quadratic function of category-specific interruption rates, scaled by period duration so that utility is homogeneous of degree one in ℓ_t :

$$u(\mathbf{x}_t, \ell_t) = \ell_t \left[\sum_{c \in \mathcal{M}} \mu_c \frac{x_{ct}(\mathbf{x}_t)}{\ell_t} + \sum_{c \in \mathcal{M}^*} \kappa_c \frac{x_{ct}^{ex}}{\ell_t} + \frac{1}{2} \sum_{c, c' \in \mathcal{M}} \Omega_{cc'} \frac{x_{ct}(\mathbf{x}_t)}{\ell_t} \frac{x_{c't}(\mathbf{x}_t)}{\ell_t} \right]. \quad (17)$$

The vector $\boldsymbol{\mu} = (\mu_{SM}, \mu_{CM}, \mu_{OT})$ captures the base marginal utility of category-specific phone activity, regardless of whether the check is externally triggered or self-initiated. The external-premium vector $\boldsymbol{\kappa} = (\kappa_{SM}, \kappa_{CM})$ captures the additional utility from acting on an unsolicited notification in the same category. Because μ_c already loads on total category- c interruptions, κ_c is the prompt or salience premium relative to an internal check.

The symmetric matrix $\boldsymbol{\Omega}$ captures within-period substitution and complementarity across app categories. Negative off-diagonal elements imply that interruptions in two categories are substitutes within a period, while the diagonal elements govern how marginal utility changes with own-category interruption intensity on the finite action support. Homogeneity in ℓ_t makes idle periods and delivery trips of different lengths directly comparable.

Continuation probability. I treat the continuation probability in equation (4) as a reduced-form function of completed cumulative spell time and observable shifters, rather than as a stopping problem solved jointly with phone-use preferences. Specifically, after a delivery trip, the probability that worker i in spell s continues at completed cumulative spell time L is

$$\rho_{is}(L) = \Lambda \left(\beta_L L + \beta_\omega \omega_{is}^{net} + \boldsymbol{\gamma}' \mathbf{w}_{is} + \eta_i + \eta_{d(i,s)} \right), \quad (18)$$

where $\Lambda(\cdot)$ is the logistic CDF, ω_{is}^{net} is net payoff per mile, $\mathbf{w}_{is}$ includes weather controls, and η_i and $\eta_{d(i,s)}$ are worker and calendar-day fixed effects. Within-spell fatigue enters the dynamic

problem through completed cumulative spell time L . The payoff and weather shifters capture observable variation in continuation incentives, while the fixed effects absorb persistent worker heterogeneity and common daily shocks.

Treating $\rho_{is}(\cdot)$ as a fitted primitive has two advantages. First, the dynamic estimation step does not need to solve a coupled stopping problem at every candidate preference parameter, which keeps estimation computationally feasible. Second, the effect of a within-period action on future values is summarized by its effect on trip duration, completed spell time, and the fitted continuation probability at that state.

Shocks. The action-specific shocks ϵ_x in equations (2) and (4) are i.i.d. Type I extreme value, independent across actions and periods, with location zero and scale normalized to one. This normalization fixes the utility scale. Under this distributional assumption, idle-period and trip-period choices have conditional logit probabilities, and the ex ante values EW and EV admit log-sum representations over the finite action set. [Appendix D.1](#) provides the full parameterized likelihood.

6.3 Estimation and identification

The preference parameters for demographic group g are

$$\Theta_g = \{\mu_g, \kappa_g, \Omega_g, \rho_g(\cdot), \alpha_g\}.$$

I estimate them sequentially. Step 1 uses idle-period choices to recover the static phone-utility parameters, $(\mu_g, \kappa_g, \Omega_g)$. Step 2 estimates the reduced-form continuation probability, $\rho_g(\cdot)$, from observed continuation decisions. Step 3 holds the first two steps fixed and estimates the revealed-preference output weight, α_g , from delivery-trip choices. The sequence follows the timing of the model: idle-period choices reveal phone utility when phone use has no production consequence, while trip-period choices reveal how workers trade off that utility against the dynamic output loss induced by interruptions. This separation parallels standard dynamic discrete-choice estimators that distinguish transition objects from choice payoffs ([Rust, 1987](#); [Hotz and Miller, 1993](#)).

A common latent-data problem appears in both the idle-period and trip-period likelihoods. The data record category-level interruption totals, $(x_{SM,t}, x_{CM,t}, x_{OT,t})$, but do not observe the period-level notification vector, $(n_{SM,t}, n_{CM,t})$, nor the decomposition of social-media and communication interruptions into external and internal components. For $c \in \mathcal{M}^*$, the observed total satisfies $x_{ct} = x_{ct}^{ex} + x_{ct}^{in}$. For OT , all observed interruptions are treated as internal. This restriction does not remove their production effect: OT interruptions still enter total interruptions $X(\mathbf{x}_t)$. It only prevents them from being attributed to the regulated external-notification supply.

Given a notification count n_c and observed total x_{ct} , the feasible set of external counts in category $c \in \mathcal{M}^*$ is

$$\mathcal{S}(n_c, x_{ct}) = \{z \in \mathbb{N} : 0 \leq z \leq \min\{n_c, x_{ct}\}\}. \quad (19)$$

For any feasible pair $(x_{SM}^{ex}, x_{CM}^{ex})$, define $\mathbf{x}(x_{SM}^{ex}, x_{CM}^{ex}; x_{SM,t}, x_{CM,t}, x_{OT,t})$ as the action vector with latent external counts $(x_{SM}^{ex}, x_{CM}^{ex})$, latent internal counts $(x_{SM,t} - x_{SM}^{ex}, x_{CM,t} - x_{CM}^{ex}, x_{OT,t})$, and observed total category counts. The likelihood contributions below integrate over both latent notification counts and feasible external/internal decompositions.

Step 1: Static likelihood from idle periods. During an idle period, duration ℓ_t and notification vector $\mathbf{n}_t$ are exogenous to the worker's action, and the continuation value $EV(L_t + \ell_t)$ does not vary with $\mathbf{x}$. The idle-period action therefore maximizes flow utility alone. Conditional on $\mathbf{n}_t$, the choice probability is

$$P^W(\mathbf{x} \mid \ell_t, \mathbf{n}_t; \boldsymbol{\mu}, \boldsymbol{\kappa}, \boldsymbol{\Omega}) = \frac{\exp(u(\mathbf{x}, \ell_t))}{\sum_{\mathbf{x}' \in \mathcal{A}(\mathbf{n}_t)} \exp(u(\mathbf{x}', \ell_t))}. \quad (20)$$

The idle-period likelihood marginalizes over latent notification counts and external/internal splits:

$$\mathcal{L}_{ist}^W(\boldsymbol{\mu}, \boldsymbol{\kappa}, \boldsymbol{\Omega}) = \sum_{n_{SM}=0}^{\bar{n}} \sum_{n_{CM}=0}^{\bar{n}} \left[\prod_{c \in \mathcal{M}^*} p(n_c; \hat{\lambda}_{is}^c \ell_t) \right] Q_{ist}^W(\mathbf{n}; \boldsymbol{\mu}, \boldsymbol{\kappa}, \boldsymbol{\Omega}), \quad (21)$$

where $p(n; \zeta)$ is the Poisson probability for n with rate ζ . The inner term is

$$Q_{ist}^W(\mathbf{n}; \boldsymbol{\mu}, \boldsymbol{\kappa}, \boldsymbol{\Omega}) = \sum_{\substack{z_{SM} \in \mathcal{S}(n_{SM}, x_{SM,t}) \\ z_{CM} \in \mathcal{S}(n_{CM}, x_{CM,t})}} P^W(\mathbf{x}(z_{SM}, z_{CM}; x_{SM,t}, x_{CM,t}, x_{OT,t}) \mid \ell_t, \mathbf{n}; \boldsymbol{\mu}, \boldsymbol{\kappa}, \boldsymbol{\Omega}).$$

The group log-likelihood is

$$L_g^W(\boldsymbol{\mu}, \boldsymbol{\kappa}, \boldsymbol{\Omega}) = \sum_{(i,s,t) \in \mathcal{T}_g^W} \log \mathcal{L}_{ist}^W(\boldsymbol{\mu}, \boldsymbol{\kappa}, \boldsymbol{\Omega}),$$

where $\mathcal{T}_g^W$ is the set of idle periods for workers in group g . The static-utility estimator is

$$(\hat{\boldsymbol{\mu}}_g, \hat{\boldsymbol{\kappa}}_g, \hat{\boldsymbol{\Omega}}_g) = \arg \max_{\boldsymbol{\mu}, \boldsymbol{\kappa}, \boldsymbol{\Omega}} L_g^W(\boldsymbol{\mu}, \boldsymbol{\kappa}, \boldsymbol{\Omega}).$$

The parameters are disciplined by three margins. Cross-category interruption frequencies identify the base utility vector $\boldsymbol{\mu}$. Variation in category-specific arrival rates, $\hat{\lambda}_{is}^{SM}$ and $\hat{\lambda}_{is}^{CM}$, shifts the posterior weight placed on external versus internal interruptions and identifies the external premia κ_{SM} and κ_{CM} . The joint distribution of category-level interruptions identifies the substitution matrix $\boldsymbol{\Omega}$ through cross-category co-movement in choices. This step is static because the continuation term is common to all feasible actions in an idle period and cancels from the conditional choice probability.

Step 2: reduced-form continuation probability. I estimate the continuation probability in equation (18) from observed spell-continuation decisions. The estimation sample consists of delivery-trip transition rows, with outcome $d_{ist} = 1$ if the worker continues into another idle-delivery cycle

after completing trip t , and $d_{ist} = 0$ if the spell ends. I estimate the logit separately by demographic group. The fitted function $\hat{\rho}_{is}(L)$ then enters Step 3 as a known transition primitive.

The identifying restriction is that, conditional on completed cumulative spell time, net payoff, weather, and the fixed effects, the continuation shock is independent of the within-period phone-use shock. This restriction corresponds to the timing in Section 3: phone-use shocks determine the current-period interruption choice, the resulting duration determines completed spell time, and the continuation decision is evaluated at that completed state.

Step 3: dynamic likelihood for the output weight. Holding $(\hat{\mu}_g, \hat{\kappa}_g, \hat{\Omega}_g, \hat{\rho}_{is})$ fixed, I estimate α_g from delivery-trip choices. The relevant state contains cumulative spell time L , trip distance m , net payoff, category-specific notification supply, and the production-duration shifter. I evaluate the value functions $EW_g(\cdot; \alpha)$ and $EV_g(\cdot; \alpha)$ on a discretized state grid. At each candidate α , value-function iteration solves the Bellman recursions using the fitted production-duration function, the fitted continuation probability, and the empirical distributions of trip miles, idle durations, and notification counts.

Conditional on notification vector $\mathbf{n}$, the trip-side deterministic payoff from action $\mathbf{x}$ is

$$R_{\mathbf{x},ist}(\alpha) = u(\mathbf{x}, \hat{\ell}_{\mathbf{x}}) + \alpha \omega_{is} m_t + \hat{\rho}_{is}(L_t + \hat{\ell}_{\mathbf{x}}) EW_{is}(L_t + \hat{\ell}_{\mathbf{x}}; \alpha), \quad (22)$$

where the action-implied trip duration is $\hat{\ell}_{\mathbf{x}} = m_t \hat{\Delta}_{g,is}(X(\mathbf{x}), m_t, L_t)$. The corresponding logit choice probability is

$$P_{ist}^V(\mathbf{x} \mid L_t, m_t, \mathbf{n}; \alpha) = \frac{\exp(R_{\mathbf{x},t}(\alpha))}{\sum_{\mathbf{x}' \in \mathcal{A}(\mathbf{n})} \exp(R_{\mathbf{x}',t}(\alpha))}.$$

As in Step 1, the trip-period likelihood integrates over the latent notification vector and the feasible external/internal splits:

$$\mathcal{L}_{ist}^V(\alpha) = \sum_{n_{SM}=0}^{\bar{n}} \sum_{n_{CM}=0}^{\bar{n}} \left[\prod_{c \in \mathcal{M}^*} p(n_c; \hat{\lambda}_{is}^c \ell_t) \right] Q_{ist}^V(\mathbf{n}; \alpha), \quad (23)$$

where

$$Q_{ist}^V(\mathbf{n}; \alpha) = \sum_{\substack{z_{SM} \in \mathcal{S}(n_{SM}, x_{SM,t}) \\ z_{CM} \in \mathcal{S}(n_{CM}, x_{CM,t})}} P_{ist}^V(\mathbf{x}(z_{SM}, z_{CM}, x_{SM,t}, x_{CM,t}, x_{OT,t}) \mid L_t, m_t, \mathbf{n}; \alpha).$$

The group log-likelihood is

$$L_g^V(\alpha) = \sum_{(i,s,t) \in \mathcal{T}_g^V} \log \mathcal{L}_{ist}^V(\alpha),$$

where $\mathcal{T}_g^V$ is the set of delivery trips for workers in group g . The output-weight estimator is

$$\hat{\alpha}_g = \arg \max_{\alpha} L_g^V(\alpha).$$

Identification of α_g comes from the contrast between idle-period and delivery-trip phone use.

Step 1 recovers the utility of interruptions from idle periods, where phone activity has no production consequence. During delivery trips, the same interruption changes predicted trip duration, completed spell time, the continuation probability, and expected future production. A larger α_g makes these dynamic output losses more costly and therefore predicts a sharper reduction in trip-period interruptions relative to the idle-period baseline. Conversely, workers whose idle and trip interruption profiles are similar reveal a lower revealed-preference value of output. Inference uses a worker-clustered bootstrap that resamples panelist histories. The bootstrap propagates sampling uncertainty from the notification arrival rates, the production-duration function, the continuation probability, and the preference estimates.

6.4 Preference estimates

Table 6 reports the estimated preference parameters by demographic group. The table is organized by economic role. Panel A reports the linear flow-utility terms, separating the base utility of category-specific phone activity from the additional utility of externally prompted notifications. Panel B reports the quadratic terms that govern within-category clustering and substitution across app categories. Panel C reports the reduced-form continuation equation. Panel D reports the revealed-preference output weight, α , estimated from delivery-trip choices.

Panel A shows that external prompts are valuable relative to otherwise similar self-initiated checks. The base-utility parameters for social media and communication, μ_{SM} and μ_{CM} , are negative in every demographic group, reflecting the fact that zero or low interruption counts are common once period duration and choice shocks are taken into account. The base utility of other apps, μ_{OT} , is less negative, consistent with other-app checks being more frequently self-initiated and less tied to external prompts. In contrast, the external-premium parameters, κ_{SM} and κ_{CM} , are positive across groups. Thus, an incoming notification shifts the utility of a check in two ways: it expands the feasible set and adds prompt value relative to an internal check in the same category.

The external premia also vary across categories and demographic groups. Social-media notification premia are large for all groups and are especially high among older workers: $\hat{\kappa}_{SM}$ reaches 1.31 for women aged 40 or older and 1.43 for men aged 40 or older. Communication notification premia are also positive, but the age profile is different. They are largest for younger workers and generally smaller for middle-aged and older workers. Thus, the high social-media notification arrival rates for young workers documented in Section 4.2 do not mechanically imply that young workers have the highest social-media prompt premium. The structural model separates exposure to notifications from the utility of acting on them.

Panel B shows two robust features of phone-use preferences. First, social-media and communication interruptions exhibit within-category reinforcement. The own-category terms $\Omega_{SM,SM}$ and $\Omega_{CM,CM}$ are mostly positive, especially for social media. On the finite action support, this means that once a worker is already engaging with social media or communication in a period, the marginal utility of another interruption in the same category rises. This pattern is consistent with clustered use: one social-media notification may lead to additional scrolling or checking

Table 6: Estimated Preference Parameters by Demographic Group

Parameter	Female			Male		
	18–29	30–39	40+	18–29	30–39	40+
Panel A. Flow utility						
Base utility: social media, μ_{SM}	–1.143 (0.085)	–1.197 (0.135)	–1.642 (0.192)	–0.887 (0.105)	–1.156 (0.097)	–1.688 (0.124)
Base utility: communication, μ_{CM}	–1.123 (0.059)	–0.927 (0.041)	–0.949 (0.052)	–1.165 (0.070)	–0.926 (0.082)	–1.037 (0.046)
Base utility: other apps, μ_{OT}	–0.409 (0.036)	–0.433 (0.024)	–0.515 (0.033)	–0.363 (0.045)	–0.369 (0.036)	–0.469 (0.035)
External premium: social media, κ_{SM}	0.870 (0.128)	0.695 (0.279)	1.305 (0.342)	0.655 (0.142)	1.026 (0.129)	1.427 (0.192)
External premium: communication, κ_{CM}	0.929 (0.119)	0.431 (0.119)	0.360 (0.167)	0.838 (0.146)	0.295 (0.397)	0.656 (0.104)
Panel B. Substitution						
Own: social media, $\Omega_{SM,SM}$	0.112 (0.015)	0.114 (0.023)	0.177 (0.027)	–0.001 (0.072)	0.102 (0.019)	0.184 (0.016)
Own: communication, $\Omega_{CM,CM}$	0.102 (0.011)	0.012 (0.033)	–0.031 (0.049)	0.086 (0.026)	0.011 (0.067)	0.057 (0.025)
Own: other apps, $\Omega_{OT,OT}$	–0.750 (0.088)	–0.658 (0.053)	–0.595 (0.061)	–0.859 (0.115)	–0.735 (0.078)	–0.666 (0.075)
Cross: social media $\times$ communication, $\Omega_{SM,CM}$	–0.904 (0.223)	–1.121 (0.169)	–0.801 (0.202)	–1.227 (0.254)	–1.782 (0.233)	–1.246 (0.255)
Cross: social media $\times$ other apps, $\Omega_{SM,OT}$	–0.746 (0.131)	–0.896 (0.136)	–0.633 (0.254)	–0.975 (0.216)	–1.124 (0.187)	–0.465 (0.159)
Cross: communication $\times$ other apps, $\Omega_{CM,OT}$	–0.764 (0.141)	–0.964 (0.067)	–0.731 (0.064)	–1.097 (0.152)	–0.984 (0.096)	–0.768 (0.121)
Panel C. Continuation probability						
Cumulative spell time, β_L	–0.0048 (0.0002)	–0.0039 (0.0002)	–0.0036 (0.0002)	–0.0044 (0.0003)	–0.0036 (0.0002)	–0.0037 (0.0002)
Net payoff per mile, β_ω	–0.074 (0.274)	0.074 (0.183)	–0.055 (0.113)	–0.132 (0.214)	0.037 (0.161)	0.170 (0.114)
Panel D. Output valuation						
Output weight, α	3.473 (0.161)	2.712 (0.153)	2.430 (0.154)	3.340 (0.245)	2.873 (0.214)	2.228 (0.165)

Notes: The table reports structural preference estimates by gender–age group. Panel A reports base utility and external-notification premia from the idle-period likelihood. Panel B reports the unique elements of the symmetric quadratic utility matrix; these curvature and substitution terms are interpreted on the finite action support used in estimation. Panel C reports the fitted continuation equation. Panel D reports the revealed-preference output weight. Standard errors are clustered by worker and reported in parentheses. The utility scale is normalized by the Type-I extreme-value shock variance.

within the same app category, and one communication prompt may lead to a sequence of related replies or message checks. This reinforcement is important for productivity because an external notification can initiate a cluster of subsequent interruptions, including internally initiated checks, rather than generating only a single isolated action.

Second, app categories substitute for one another within a period. All cross-category terms are negative: social media and communication are substitutes, social media and other apps are substitutes, and communication and other apps are substitutes. This implies that attention is scarce across categories. When activity in one category is high, the marginal utility of switching into another category falls. The substitution estimates are central for counterfactuals: restricting social-media notifications need not reduce total phone activity one-for-one if workers substitute toward communication or other apps.

Panel C reports the fitted continuation equation. The coefficient on cumulative spell time, β_L , is negative and precisely estimated in every demographic group. This supports the reduced-form fatigue channel in the dynamic model: as completed spell time rises, workers are less likely to continue into another delivery cycle. The net-payoff coefficient, β_ω , is smaller and less stable across groups, suggesting that high-frequency continuation decisions are more strongly related to elapsed spell time and worker/day heterogeneity than to the measured variation in net payoff per mile. These continuation estimates are not preferences over phone use; they determine how an interruption-induced delay changes future work opportunities.

Panel D reports the output weight, α , the key dynamic preference parameter. The estimates are positive in every group, implying that workers discipline phone use during delivery trips when interruptions carry production consequences. The age gradient is pronounced: $\hat{\alpha}$ is highest for workers aged 18–29 and declines with age for both women and men. Among women, the estimate falls from 3.47 for ages 18–29 to 2.43 for ages 40 and older; among men, it falls from 3.34 to 2.23. These estimates indicate that younger workers reveal a higher value of output while on the job, even though they also face higher social-media notification supply.

Taken together, the estimates show that notifications matter through both supply and preferences. External prompts carry positive premia, social-media and communication use exhibit within-category clustering, and app categories substitute for one another within a period. At the same time, workers reduce phone use during delivery trips because interruptions reduce expected output through the production-duration and continuation channels. These estimated primitives are the behavioral inputs for the counterfactual policies in Section 7.

7 Platform Regulation

The previous sections show that notifications affect production through two margins. First, notification supply expands the worker’s feasible set of externally triggered interruptions. Second, interruptions lengthen delivery trips, raise cumulative spell time, and reduce expected output. This section combines the estimated notification process, production-duration function, and pref-

erence parameters to evaluate policies that restrict the timing, targeting, or availability of notifications during delivery work.

The analysis focuses on productivity. Section 3.2 shows that, relative to the revealed-preference benchmark, the welfare correction associated with planner or regulator wedges is proportional to expected output. I therefore report counterfactual effects on two productivity outcomes: minutes per delivery trip and miles completed per spell hour. The first outcome captures the direct trip-duration channel; the second captures the dynamic output channel, because shorter trips free time for additional production within a working spell.

7.1 Counterfactual design

The baseline is the estimated model under the observed notification environment. For each counterfactual policy, I hold fixed the estimated preferences, production-duration function, net payoff, weather controls, and spell composition, and change either the worker’s feasible action set or the distribution of external notifications. Workers then re-optimize under the new environment. This procedure isolates the productivity consequences of changing notification supply or access while keeping the underlying production technology and preference parameters fixed.

I report worker-level gains relative to the baseline using two productivity metrics: minutes per delivery trip and miles completed per spell hour. The first metric captures the direct trip-duration channel. The second captures the output channel: shorter trips raise miles completed per unit of working time, both by reducing time spent on a given delivery and by increasing the scope for additional deliveries within a spell. I compute counterfactual outcomes at the worker level and then summarize the distribution of gains within demographic groups.

Table 7 defines the counterfactual policies. I organize them into three groups. The first is an upper-bound benchmark that eliminates all non-work interruptions and measures the total productivity cost of interruptions. The second group consists of user-side notification tools, such as do-not-disturb or app-specific muting. These policies do not change the platform’s underlying notification technology, but they remove visible external prompts from the worker’s choice set during work. The third group consists of supply-side platform policies that restrict the timing or targeting of external prompts before the worker makes a phone-use decision, including batching rules and restrictions on personalization.

7.2 Productivity gains

Figure 5 reports the main counterfactual results. Panel A reports the worker-level reduction in delivery-trip duration, measured in percent relative to the baseline. Panel B converts the output gain into extra income per hour, computed as additional miles completed per spell hour valued at the worker’s net payoff per mile. Positive values indicate productivity gains in both panels. The top row reports the upper-bound benchmark that removes all interruptions. The remaining rows report feasible notification policies that mute, batch, or depersonalize external notifications.

Table 7: Counterfactual Policies

Policy	Model intervention	Interpretation
I. Upper-bound benchmark		
Ban all interruptions	Eliminate all non-work interruptions during working spells.	Measures the total productivity cost of non-work phone interruptions.
II. User-side tools		
Mute notifications: SM	Set visible SM external notification arrivals to zero.	Removes social-media prompts while allowing internal checks and communication notifications.
Mute notifications: CM	Set visible CM external notification arrivals to zero.	Removes communication prompts while allowing internal checks and social-media notifications.
Mute notifications: both	Set visible external notification arrivals in both categories to zero.	Removes all modeled external prompts while allowing internally initiated checks.
III. Platform policies		
Batch notifications: SM	Cap SM notification arrivals at one within each 30-minute interval.	Limits high-frequency bursts of social-media prompts.
Batch notifications: CM	Cap CM notification arrivals at one within each 30-minute interval.	Limits high-frequency bursts of communication prompts.
Batch notifications: both	Cap total SM+CM notification arrivals at one within each 30-minute interval.	Limits the combined external prompt load across both categories.
No personalization: SM	Replace personalized SM notification arrival rates with the app-level median.	Removes persistent user-specific targeting in social-media notifications.

Notes: The table defines the counterfactual policies used in the simulations. SM denotes social media and CM denotes communication, the two external notification-generating categories in the structural model. User-side notification tools remove visible prompts from the worker's choice set during work but leave internally initiated checks available. Supply-side platform policies perturb the distribution of external notification arrivals. The first row is an upper-bound benchmark that removes all non-work interruptions, including internally initiated checks.

The first message is that interruptions impose large productivity costs. Eliminating all non-work interruptions reduces delivery-trip duration by 12.0 percent on average, or about 3.8 minutes per delivery trip. The corresponding output gain is \$1.43 per hour, which equals \$2,578 on a full-time-equivalent annual basis. The density in the top row is wide, indicating substantial heterogeneity in the cost of interruptions. Most workers experience meaningful productivity losses, and a nontrivial upper tail faces much larger losses than the mean.

The second message is that external notifications explain an important, but incomplete, share of the total interruption cost. Muting both social-media and communication notifications reduces trip duration by 2.3 percent and raises income by \$0.27 per hour, or \$485 annually on the same full-time-equivalent basis. This is about one-fifth of the full no-interruption benchmark. Muting social-media notifications alone and muting communication notifications alone have similar average effects: each reduces trip duration by about 1.2 percent and raises income by about \$0.14 per hour, corresponding to annualized gains of \$254 and \$246, respectively. The gap between muting both notification categories and eliminating all interruptions is informative: external prompts matter, but internally initiated checks and other-app interruptions also account for a large share of productivity losses.

The third message is that policy design matters. Batching notifications preserves some notifi-

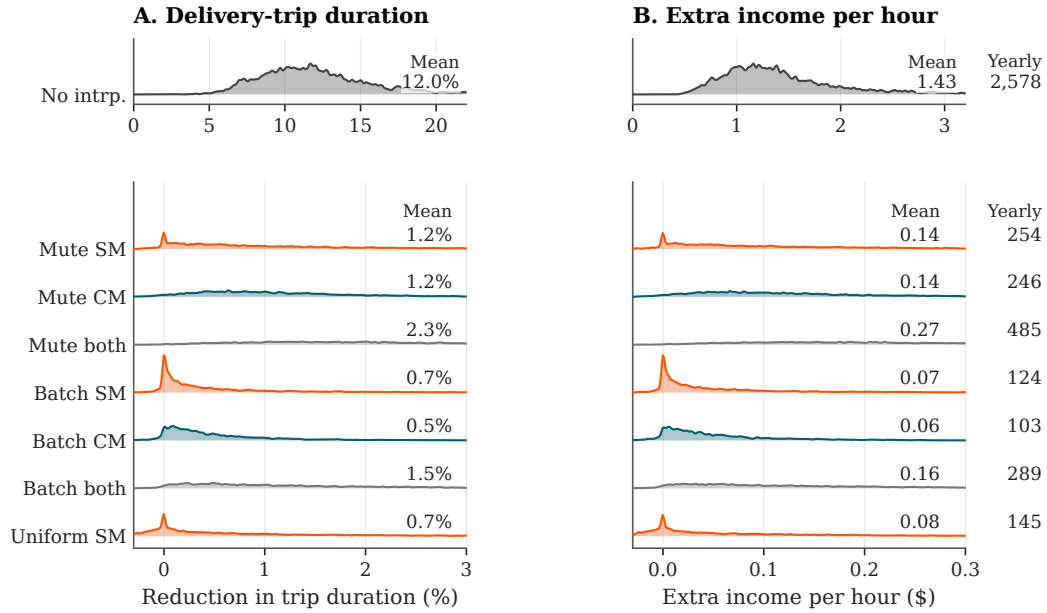


Figure 5: Productivity gains: Duration reduction and extra income

Notes: The figure reports worker-level counterfactual gains relative to the baseline notification environment. Panel A reports the percent reduction in minutes per delivery trip. Panel B reports extra income per hour, computed from additional miles per spell hour valued at the worker’s net payoff per mile. Each row plots the worker-level density for one policy. The “Mean” column reports worker-spell-weighted means. In Panel B, the “Yearly” column annualizes the mean extra income per hour by multiplying it by 1,800 hours; this is a full-time-equivalent scaling used for interpretation, not an assumption about annual hours worked by couriers in the sample. Policies are defined in Table 7. SM denotes social media and CM denotes communication.

cation flow while limiting high-frequency bursts. Batching both categories reduces trip duration by 1.5 percent and raises income by \$0.16 per hour, or \$289 annually. This captures a substantial share of the gain from muting both categories while imposing a weaker restriction. Category-specific batching has smaller effects: batching social-media notifications reduces duration by 0.7 percent and raises income by \$0.07 per hour, while batching communication notifications reduces duration by 0.5 percent and raises income by \$0.06 per hour. These results suggest that notification burstiness is an important component of the productivity cost of external prompts.

Depersonalizing social-media notifications also improves productivity, but the average effect is smaller. Replacing personalized social-media notification supply with a uniform benchmark reduces trip duration by 0.7 percent and raises income by \$0.08 per hour, or \$145 annually. Unlike muting, this policy does not eliminate social-media prompts. Instead, it compresses the cross-worker dispersion in notification supply. Its distribution therefore has a thin negative tail: workers who receive low social-media notification supply at baseline may face more prompts under depersonalization. The positive mean reflects larger gains among highly targeted workers.

The distributions in Figure 5 are as important as the means. For broad restrictions, the densities have long right tails, showing that average effects mask a subset of workers with much larger gains. For narrower policies, such as batching or uniformizing social-media prompts, the mass is closer to zero but still has a visible upper tail. This pattern reflects the interaction of three estimated

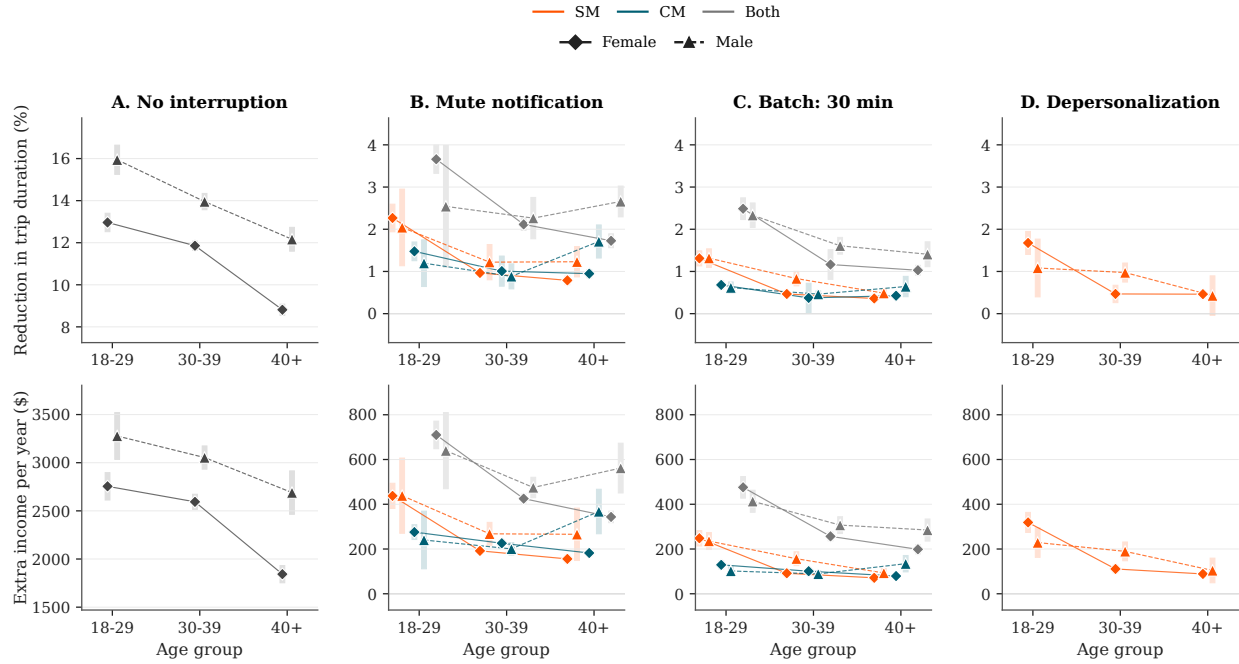


Figure 6: Counterfactual productivity gains by demographic group

Notes: The figure reports demographic-group counterfactual gains relative to the baseline notification environment. The top row reports the percent reduction in delivery-trip duration; the bottom row reports extra income per year, computed as extra income per hour multiplied by 1,800 hours. Points are worker-spell-weighted means by gender-age group. Translucent rectangular bars report 95 percent confidence intervals for the worker-level mean, treating workers as the independent units. Diamond markers with solid lines denote women, and triangle markers with dashed lines denote men. Orange denotes social-media policies, teal denotes communication policies, gray denotes both-category policies, and dark gray denotes the no-interruption benchmark. The no-interruption column uses its own y-axis; the other three columns share a common y-axis within each row.

objects: heterogeneous notification exposure, heterogeneous production costs of interruptions, and heterogeneous preferences over phone activity. The next subsection examines how these gains vary across demographic groups and how that incidence maps to the notification and production heterogeneity documented earlier.

7.3 Distributional effects and policy interpretation

The productivity gains in Figure 5 mask substantial heterogeneity across workers. Figure 6 summarizes this incidence by gender and age. The top row reports reductions in delivery-trip duration; the bottom row converts the same output gains into full-time-equivalent annual income, using 1,800 hours. The four columns separate the upper-bound no-interruption benchmark, mute policies, batching policies, and social-media depersonalization.

The no-interruption benchmark shows that the total productivity cost of interruptions is largest for younger workers and for men. Eliminating all non-work interruptions reduces delivery-trip duration by 15.9 percent for men aged 18–29 and 13.0 percent for women aged 18–29. The gains decline with age, reaching 12.2 percent for men aged 40 or older and 8.8 percent for women aged

40 or older. The same ranking appears in income units. The annualized gain is \$3,278 for men aged 18–29, \$2,755 for women aged 18–29, \$2,690 for men aged 40 or older, and \$1,843 for women aged 40 or older. These differences combine the two forms of heterogeneity documented earlier: younger workers face higher social-media notification exposure, while men face larger production losses from a given interruption.

Among feasible notification policies, the incidence depends on which source of external notifications is restricted. Muting social-media notifications has the largest effects for young workers. It reduces trip duration by 2.3 percent for women aged 18–29 and 2.0 percent for men aged 18–29, compared with 0.8 percent for women aged 40 or older and 1.2 percent for men aged 40 or older. In annualized income terms, the gains from muting social-media notifications are \$437 for women aged 18–29 and \$438 for men aged 18–29, compared with \$156 and \$266 for women and men aged 40 or older. This age gradient mirrors the notification estimates in Section 4.2: social-media notification supply is highest among younger workers.

Communication restrictions have a different incidence. Muting communication notifications produces more balanced gains across age groups and is especially relevant for older men. For men aged 40 or older, muting communication notifications reduces trip duration by 1.7 percent and raises annualized income by \$368. For women aged 18–29, the corresponding gains are 1.5 percent and \$276. Thus, communication policies do not primarily target the young-worker social-media margin; they affect a broader and somewhat older set of workers, consistent with the flatter and older-tilted communication arrival rates documented in Section 6.1.

Policies that target both notification categories combine these margins and generate the largest feasible gains. Muting both social-media and communication notifications reduces trip duration by 3.7 percent for women aged 18–29 and 2.5 percent for men aged 18–29. The annualized gains are \$710 and \$640, respectively. The effect is smaller but still economically meaningful for older workers: \$343 for women aged 40 or older and \$562 for men aged 40 or older. These results show why category-specific policies can miss an important part of the productivity cost. Workers receive external prompts from both app families, and the relevant policy burden depends on the joint notification environment.

Batching policies attenuate the same patterns. Batching both categories reduces trip duration by 2.5 percent for women aged 18–29 and 2.3 percent for men aged 18–29, with annualized gains of \$476 and \$413. The corresponding gains for workers aged 40 or older are smaller, equal to \$199 for women and \$285 for men. Category-specific batching has more modest effects: batching social-media notifications mainly benefits younger workers, while batching communication notifications is more evenly distributed and relatively more useful for older men. Batching therefore provides an intermediate policy: it preserves some notification flow but limits high-frequency bursts that generate clusters of interruptions.

Depersonalizing social-media notifications has the most targeted incidence. It reduces trip duration by 1.7 percent for women aged 18–29 and 1.1 percent for men aged 18–29, corresponding to annualized gains of \$319 and \$229. The effect declines sharply with age, reaching \$89 for women aged 40 or older and \$104 for men aged 40 or older. This pattern is natural because the policy

compresses personalized social-media notification supply toward a uniform benchmark rather than eliminating social-media prompts. Its gains are concentrated among workers who are highly targeted at baseline, especially younger workers.

Figure 6 therefore highlights the policy trade-off between magnitude and precision. The no-interruption benchmark generates the largest gains but removes all non-work phone activity, including internally initiated checks that workers may value. Muting policies are narrower: they remove external prompts in targeted categories while preserving internal checks. Batching policies are narrower still, reducing bursts while preserving some notification flow. Depersonalization is the most targeted policy, changing the allocation of social-media prompts across workers rather than eliminating the category.

These distinctions matter for regulation. User-side tools, such as muting or do-not-disturb rules, require workers to adopt and maintain commitment devices during work. Supply-side rules act before the worker faces the prompt, limiting the arrival, timing, or targeting of external interruptions. This distinction is important when the workers most exposed to notifications are also those for whom interruptions are most costly. At the same time, communication notifications may carry coordination value for the worker or for others. The estimates should therefore be read as the productivity side of the trade-off, not as a blanket case for eliminating communication prompts. Finally, the reported gains do not monetize consumer waiting costs, courier congestion externalities, or safety costs from the catch-up response documented in Section 5.3. Incorporating those externalities would raise the social value of policies that reduce in-trip interruptions.

8 Conclusion

This paper shows that notification design is a productivity-relevant margin of digital platform policy. Push notifications are often evaluated by their ability to generate engagement, but the same prompts can arrive when attention has a high opportunity cost. External notifications expand the feasible set of immediate phone engagement, raise the value of responding to prompts, generate clustered interruptions, and slow production.

The policy implication is not that notifications should be eliminated. Communication prompts may convey coordination value, and workers may value internally initiated phone use. The results instead point to a design trade-off. Broad muting tools generate the largest feasible productivity gains among the notification policies considered, while batching preserves some information flow and captures a meaningful share of the gains by limiting high-frequency bursts. Depersonalizing social-media notifications has smaller average effects but targets a distinct concern: the unequal allocation of prompt exposure across users. Because gains are largest for workers who face both high notification exposure and high production costs of interruptions, notification policy also has distributional consequences. Design choices about timing, personalization, and category-specific prompting therefore affect not only engagement, but also worker output, earnings, and the incidence of attention costs.

The analysis leaves several directions for future work. First, this paper quantifies the productivity side of the trade-off; a fuller welfare analysis would also measure the benefits of notifications to workers, platforms, consumers, and social contacts, as well as safety, waiting-time, and congestion externalities. Second, food delivery is a setting where output is unusually observable and the smartphone is both a work tool and an interruption device. Similar measurement strategies could be applied to other forms of mobile or computer-mediated work, where output is less directly observed but attention costs may be just as important. Third, field experiments that randomize notification timing, batching, personalization, or work-mode defaults would provide direct evidence on adaptation, substitution across apps, and equilibrium responses by users and platforms. More broadly, the paper highlights the need to study digital prompts as part of an attention-allocation system. Platform messages create value by reaching users at the right moment, but the social value of that moment depends on what else the user is trying to do.

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